第八章 多元函数微分学

第一节 多元函数的极限与连续性

§1.1 多元函数的概念

A,B两个非空实数积集合 $\{(x,y): x\in A, y\in B\}$ 称为A与B的直积

记作 $A\times B$，特别地 $A\times A\triangleq A^2$。

P.62 定义 8.2 设n个R的直积

$$\underbrace{R\times R\times\ldots\times R}_{n个}\triangleq R^n=\{(x_1,x_2,\ldots,x_n): x_i\in R, i=1,2,\ldots n.\}$$

（称为R^n为n维欧氏空间），且$D\subseteq R^n$是非空集合。

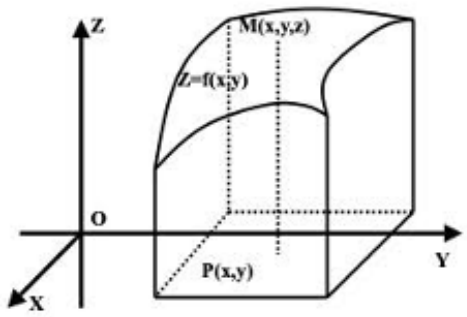
若存在一个对应法则f，对每一个$P(x_1,x_2,\ldots,x_n)\in D$,都有唯一的实数$u$与之对应，则称$f$是$D$上的$n$元函数，记作$u=f(P)=f(x_1,x_2,\ldots,x_n)$

其中D称为函数f的定义域，$x_1,x_2,\ldots,x_n$称为自变量，u为因变量。

$R(f)=\{u: u=f(x_1,x_2,\ldots,x_n), P(x_1,x_2,\ldots,x_n)\in D\}$称为函数的值域.

$\{(x_1,x_2,\ldots,x_n,u): u=f(x_1,x_2,\ldots,x_n), P(x_1,x_2,\ldots,x_n)\in D\}\subset R^{n+1}$

称为$u=f(x_1,x_2,\ldots,x_n)$的图形.

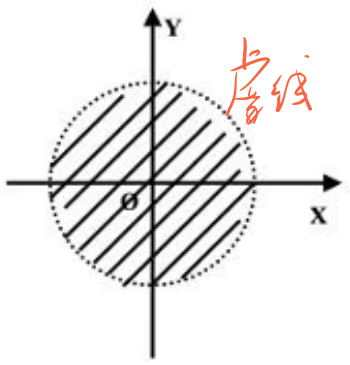


简单地，$f: D\subseteq R^n\longrightarrow R$

$(x_1,x_2,\ldots,x_n)\longrightarrow u$

$u=f(P)=f(x_1,x_2,\ldots,x_n)$

本章重点研究二元函数，所得的结果可直接推广到更多元函数上去。



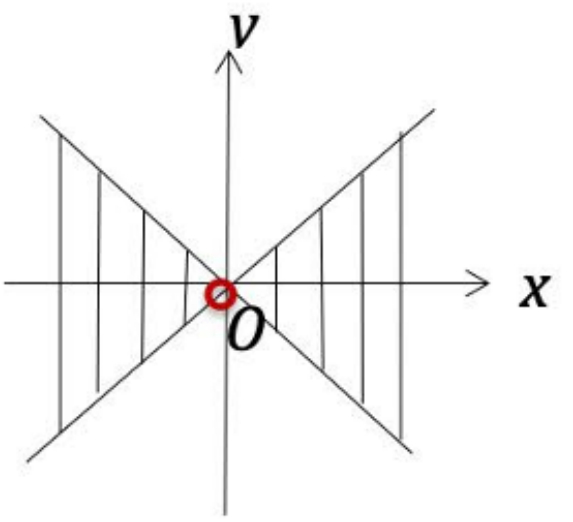
P.63 例 1 求 $z=\dfrac{1}{\sqrt{1-x^2-y^2}}$的定义域.

解 由$1-x^2-y^2>0$，即$x^2+y^2<1$.这是Oxy平面上一个以原点为中心，以1为半径的圆，包括内部不包括圆周. 注：图形用阴影表示出来,对于边界:若包括用实线,若不包括用虚线.

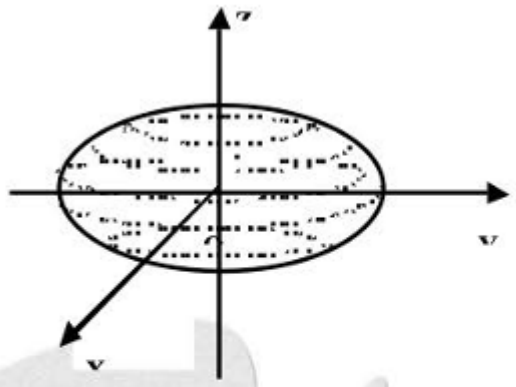
P.63 例 2 求$z=\arcsin\dfrac{y}{x}$的定义域.

解 存在域为满足不等式$\left|\dfrac{y}{x}\right|\le 1$,即$|y|\le|x|\ (x\neq 0)$.

这是一对对顶的直线阴影部分所示，不包括原点。



P.63 例 3　求$u=\ln\left(1-x^2-\frac{y^2}{4}-z^2\right)$的定义域.

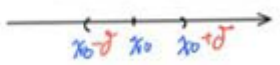
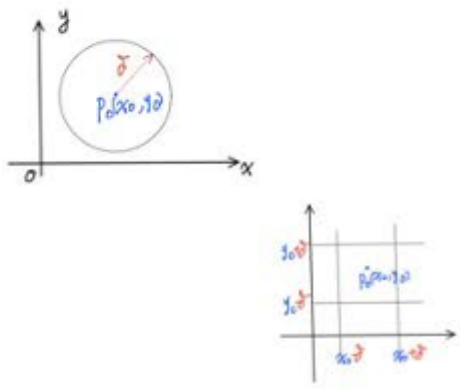
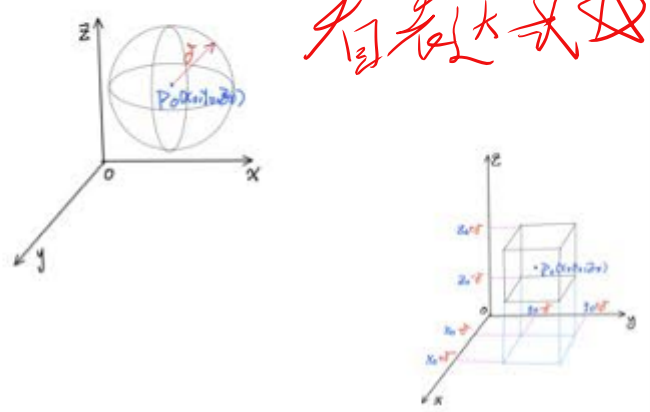


解: $1-x^2-\frac{y^2}{4}-z^2>0$, 即$x^2+\frac{y^2}{4}+z^2<1$.

因此，该定义域是$x^2+\frac{y^2}{4}+z^2=1$ 的内部, 不包括边界曲面.

§1.2　平面点集

一、邻域

	一维 $\rho(P_0,P)\triangleq\sqrt{(x-x_0)^2}$	二维 $\rho(P_0,P)\triangleq\sqrt{(x-x_0)^2+(y-y_0)^2}$	三维 $\rho(P_0,P)\triangleq$ $\sqrt{(x-x_0)^2+(y-y_0)^2+(z-z_0)^2}$
邻域	$\rho(P_0,P)<\delta$	$\rho(P_0,P)<\delta$	$\rho(P_0,P)<\delta$
空心邻域	$0<\rho(P_0,P)<\delta$	$0<\rho(P_0,P)<\delta$	$0<\rho(P_0,P)<\delta$
方邻域	$\lvert x-x_0\rvert<\delta$	$\begin{cases}\lvert x-x_0\rvert<\delta\\ \lvert y-y_0\rvert<\delta\end{cases}$	$\begin{cases}\lvert x-x_0\rvert<\delta\\ \lvert y-y_0\rvert<\delta\\ \lvert z-z_0\rvert<\delta\end{cases}$
空心方邻域	$\lvert x-x_0\rvert<\delta$且$x\neq x_0$	$\begin{cases}\lvert x-x_0\rvert<\delta\\ \lvert y-y_0\rvert<\delta\end{cases}$且$\begin{cases}x\neq x_0\\ y\neq y_0\end{cases}$	$\begin{cases}\lvert x-x_0\rvert<\delta\\ \lvert y-y_0\rvert<\delta\\ \lvert z-z_0\rvert<\delta\end{cases}$且$\begin{cases}x\neq x_0\\ y\neq y_0\\ z\neq z_0\end{cases}$
图像			

二、点和点集之间的关系:

任意点$P_0\in R^2$与任意一个点集$E\subset R^2$之间必有以下三种关系之一:

$P_0\in R^2$	内点	外点	界点
$E\subset R^2$	$P_0\in E$	$P_0\notin E$	$P_0\in E$ 或 $P_0\notin E$
$U(P_0)$	$U(P_0)\subset E$	$U(P_0)\cap E=\phi$	$U(P_0,\delta)\cap E\neq\phi$且$U(P_0,\delta)\cap E^C\neq\phi$
	$\operatorname{int}E$		∂E

$\mathring{U}(P_0,\delta)\xrightarrow{\text{def}}\{P(x,y):0<\rho(P_0,P)<\delta\}=\{P(x,y):(x-x_0)^2+(y-y_0)^2<\delta^2\}$

δ不指明 $\mathring{U}(P_0)$

$U(P_0)$

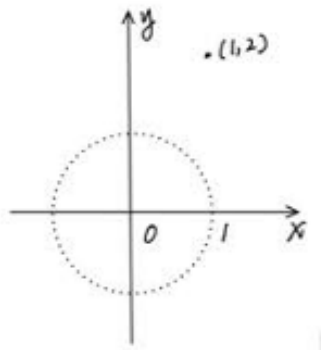
例如：

(1)平面点集$D=\{(x,y)\in R^2: x^2+y^2<1\}\cup\{(1,2)\}$

内点：$(x,y): x^2+y^2<1$

界点：$(1,2)\in D$，满足$x^2+y^2=1$的$(x,y)\notin D$

边界：$\partial D=\{(x,y)\in R^2: x^2+y^2=1\}\cup\{(1,2)\}$

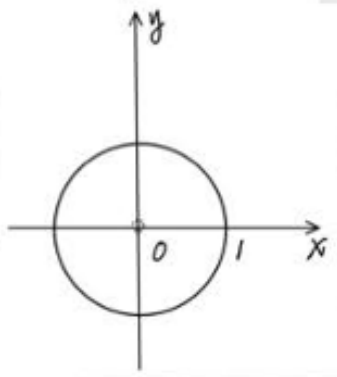


(2)平面点集$D=\{(x,y)\in R^2: 0<x^2+y^2\leq 1\}$

内点：$(x,y): 0<x^2+y^2<1$

界点：$(0,0),(x,y): x^2+y^2=1$

边界：$\partial D=\{(x,y)\in R^2: x^2+y^2=1\}\cup\{(0,0)\}$



点集

开集——$\text{int}\,E=E$，称E为开集。$(\text{int}\,\phi=\phi, \text{int}\,R^2=R^2)\Rightarrow$开集

闭集——E^C是开集，称E为闭集。$((R^2)^C=\phi,\ \phi^C=R^2)\Rightarrow$闭集

例 $P_0\in R^2, U(P_0,\delta), U^0(P_0,\delta), R^2, \phi$都是开集，$R^2, \phi$也是闭集.

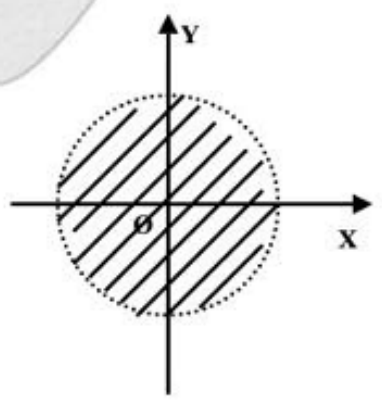
若E中任意两点之间都可用一条完全含于E的有限条折线（由有限条直线连接而成的折线）相连接，称E具有连通性.

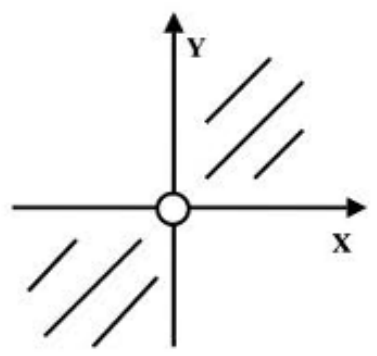
若E既是开集又具有连通性，称E为开域.

例如$\{(x,y): x^2+y^2<1\}$是一个开域。

但$\{(x,y): xy>0\}$不是一个开域。

开域连同其边界所成的点集称为闭域。





开域、闭域或者开域连同其一部分边界点组成的点集统称为区域。

设$E\subset R^2$，若存在常数$r>0$，使$E\subset U(O,r)$，称E是有界点集（有界集），否则称E是无界集。O是坐标原点，也可以是其它点。

§1.3 二元函数的极限与连续

一、极限

多元函数的极限定义：

$P_0=(x_1^0,x_2^0,\ldots,x_n^0)\in R^n, P=(x_1,x_2,\ldots,x_n)\in U^0(P_0)\subseteq R^n$

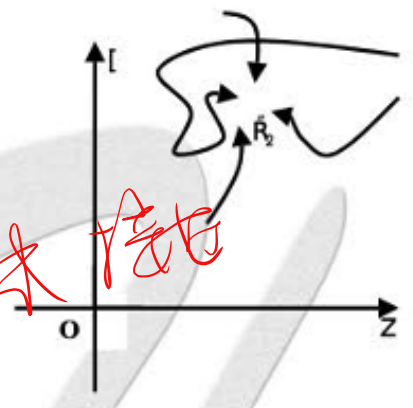
设n元函数$f(P)$在$U^0(P_0,\delta_0)$内有定义，若存在一常数A，$\forall\varepsilon>0,\exists\delta>0\ (\delta\leq\delta_0)$，当$0<$

$\rho(P,P_0)<\delta$,都有$|f(P)-A|<\varepsilon$成立，则称A为n元函数$f(P)$当$P\to P_0$时的极限，记作：

$$\lim_{P\to P_0} f(x)=A \text{ 或 } f(P)\to A\ (P\to P_0)$$

各个方向趋向

当$n=2$时，为二元函数，二元函数的极限定义见 P.65 定义 8.4。



注意：这个极限与点$P(x,y)$趋于点$P_0(x_0,y_0)$的方式无关，只要P距离P_0充分接近。点P趋于P_0点方式可有无数种，而一元函数仅有左，右两侧极限。

无数种方式接近

多元函数极限的归结原则：

若$f(P)$在$U^0(P_0,\delta_0)$内有定义，则：

$$\lim_{P\to P_0} f(P)=A \Leftrightarrow \text{对于}U^0(P_0,\delta_0)\text{的任何子集}E,\ \lim_{\substack{P\to P_0\\(P\in E)}} f(P)=A$$

同样可用归结原则，若发现点P按两个特殊的路径趋于点P_0时，$f(P)$极限存在，但不相等，则$f(P)$在该点P_0极限不存在. 这是判断多元函数极限不存在的重要方法之一.

一元函数极限中除了单调有界定理外，其余的有关性质和结论，在二元函数极限里都有相同的结果。例如：P.66 (8.1)

$$\lim_{\substack{x\to x_0\\y\to y_0}} f(x,y)=A \Leftrightarrow f(x,y)=A+\alpha\ \left(\text{其中}\lim_{\substack{x\to x_0\\y\to y_0}}\alpha=0\right)$$

求多元函数的极限，一般都是转化为一元函数的极限来求，或利用夹逼定理来计算。

补充题 1: 证明：$\lim\limits_{(x,y)\to(0,1)} \dfrac{x+2y-1}{x^2+y^2}=1$ (重点看)

要证明是有正<ε

解：不妨设：$-1\le x\le 1,\dfrac{1}{2}\le y\le\dfrac{3}{2}$

$\forall 1>\varepsilon>0,\exists\delta=\dfrac{\varepsilon}{12}$，当$|x-0|<\delta,|y-1|<\delta$且$(x,y)\neq(0,1)$时，

有正有负也为0，则等号

$$\left|\frac{x+2y-1}{x^2+y^2}-1\right|=\frac{|x^2-x+(y-1)^2|}{x^2+y^2}\le 4|x^2-x+(y-1)^2|\le 4|x|\cdot|x-1|+4(y-1)^2$$

$$\le 8|x|+4|y-1|<\varepsilon.$$

$|x-1|<2$　$|y-1|<1$

代入基本

$$\therefore \lim_{(x,y)\to(0,1)} \frac{x+2y-1}{x^2+y^2}=1$$

注意：解题规范

P.66 例 4 求 $\lim\limits_{\substack{x\to\infty\\ y\to\infty}} \dfrac{x+y}{x^2-xy+y^2}$

提示：由 $0\le\left|\dfrac{x+y}{x^2-xy+y^2}\right|\le\dfrac{|x+y|}{2|xy|-xy}\le\dfrac{|x+y|}{|xy|}\le\dfrac{1}{|x|}+\dfrac{1}{|y|}$,

$\lim\limits_{\substack{x\to\infty\\ y\to\infty}}\left(\dfrac{1}{|x|}+\dfrac{1}{|y|}\right)=0 \Rightarrow \lim\limits_{\substack{x\to\infty\\ y\to\infty}}\left|\dfrac{x+y}{x^2-xy+y^2}\right|=0,$ 夹逼

P.66 例 6 求 $\lim\limits_{\substack{x\to+\infty\\ y\to+\infty}}\left(\dfrac{xy}{x^2+y^2}\right)^{x^2}$

提示 由 $0\le\left(\dfrac{xy}{x^2+y^2}\right)^{x^2}\le\left(\dfrac{1}{2}\right)^{x^2}$，又 $\lim\limits_{x\to+\infty}\left(\dfrac{1}{2}\right)^{x^2}=0 \Rightarrow \lim\limits_{\substack{x\to+\infty\\ y\to+\infty}}\left(\dfrac{xy}{x^2+y^2}\right)^{x^2}=0$ 夹逼

补充题 2: 证明：$\lim\limits_{(x,y)\to(0,0)}\dfrac{xy^2}{x^2+y^2}=0$

解：$\because |xy|\le\dfrac{1}{2}(x^2+y^2)$, 则 $0\le\left|\dfrac{xy^2}{x^2+y^2}\right|\le\dfrac{1}{2}|y|$ 夹逼 $\Rightarrow \lim\limits_{(x,y)\to(0,0)}|y|=0 \Rightarrow \lim\limits_{(x,y)\to(0,0)}\dfrac{xy^2}{x^2+y^2}=0$

"$\varepsilon-\delta$"定义：$\forall\varepsilon>0,\exists\delta=\varepsilon$, 当 $0<x^2+y^2<\delta^2$ 时，有：

$$\left|\frac{xy^2}{x^2+y^2}\right|\le\frac{1}{2}|y|\le|y|\le\sqrt{x^2+y^2}=\delta=\varepsilon.$$

⚠ 一定要掌握！

补充题 3: 设 $f(x,y)=\begin{cases}\dfrac{xy^2}{x+y}, & (x+y\neq0)\\ 0, & (x+y=0)\end{cases}$, 求 $\lim\limits_{(x,y)\to(0,0)}f(x,y)$.

提示：$\because \lim\limits_{\substack{y=kx\\ x\to0,k\neq-1}}\dfrac{xy^2}{x+y}=\lim\limits_{\substack{y=kx\\ x\to0,k\neq-1}}\dfrac{k^2x^3}{(1+k)x}=0$, $\lim\limits_{\substack{y=x^3-x\\ x\to0}}\dfrac{xy^2}{x+y}=\lim\limits_{\substack{y=x^3-x\\ x\to0}}\dfrac{x(x^3-x)^2}{x+(x^3-x)}=1$

∴上述极限不存在。

进一步：计算：$\lim\limits_{(x,y)\to(0,0)}\dfrac{xy^{2025}}{x+y}$

解：$\because \lim\limits_{\substack{y=kx^{2026}-x\\ x\to0,k\neq0}}\dfrac{xy^{2025}}{x+y}=\lim\limits_{\substack{y=kx^{2026}-x\\ x\to0,k\neq0}}\dfrac{x(kx^{2026}-x)^{2025}}{x+(kx^{2026}-x)}=-\dfrac{1}{k}.$

∴其极限与趋向方式有关，故上述极限不存在。

补充题 4： 求 $\lim\limits_{\substack{x\to 0\\ y\to 0}}(x^2+y^2)^{x^2y^2}$.

换ln之后换元

提示：

$$(x^2+y^2)^{x^2y^2}=e^{x^2y^2\ln(x^2+y^2)} \Rightarrow 0\le|x^2y^2\ln(x^2+y^2)|\le\left|\frac{(x^2+y^2)^2}{4}\ln(x^2+y^2)\right|,$$

$$\lim_{\substack{x\to 0\\ y\to 0}}\frac{(x^2+y^2)^2}{4}\ln(x^2+y^2)\overset{令x^2+y^2=t}{=}\frac{1}{4}\lim_{t\to 0^+}\frac{\ln t}{t^{-2}}\left(\frac{\infty}{\infty}\right)=\frac{1}{4}\lim_{t\to 0^+}\frac{\frac{1}{t}}{-2t^{-3}}=0.$$

根据夹逼定理知 $\lim\limits_{\substack{x\to 0\\ y\to 0}}x^2y^2\ln(x^2+y^2)=0$，故原式 $=\lim\limits_{\substack{x\to 0\\ y\to 0}}e^{x^2y^2\ln(x^2+y^2)}=e^0=1.$

⚠ 以下例题，做作业之前好好看看！

P.67 例 7 研究函数 $f(x,y)=\begin{cases}\frac{2xy}{x^2+y^2}, & x^2+y^2\neq 0;\\ 0, & x^2+y^2=0\end{cases}$ 在点(0,0)处极限是否存在.

解 由 $\lim\limits_{\substack{x\to 0\\ y=kx}}\frac{2x\cdot kx}{x^2+k^2x^2}=\frac{2k}{1+k^2}$，知 $\lim\limits_{\substack{x\to 0\\ y\to 0}}f(x,y)$ 不存在

y=kx 换元

类似地： $\lim\limits_{\substack{x=ky^2\\ y\to 0}}\frac{xy^2}{x^2+y^4}=\frac{k}{k^2+1}$　$\lim\limits_{\substack{x=ky^3\\ y\to 0}}\frac{xy}{x^2+y^4}=k$

二、累次极限与二重极限：

加时说：高阶无穷小

累次极限 $\lim\limits_{x\to x_0}\lim\limits_{y\to y_0}f(x,y)$，$\lim\limits_{y\to y_0}\lim\limits_{x\to x_0}f(x,y)$ 与二重极限 $\lim\limits_{\substack{x\to x_0\\ y\to y_0}}f(x,y)$

⚠ 累次极限本质上是二次求一元函数的极限，二重极限是二元函数的极限。

P.67 例 7 研究函数 $f(x,y)=\begin{cases}\frac{2xy}{x^2+y^2}, & x^2+y^2\neq 0;\\ 0, & x^2+y^2=0\end{cases}$ 在点(0,0)处极限是否存在.

解： $\lim\limits_{\substack{y=kx\\ x\to 0}}\frac{2xy}{x^2+y^2}=\lim\limits_{\substack{y=kx\\ x\to 0}}\frac{2kx^2}{(k^2+1)x^2}=\frac{2k}{k^2+1}\Rightarrow\lim\limits_{\substack{x\to 0\\ y\to 0}}\frac{2xy}{x^2+y^2}$ 不存在

而 $\lim\limits_{y\to 0}\lim\limits_{x\to 0}\frac{2xy}{x^2+y^2}=\lim\limits_{y\to 0}\frac{0}{y^2}=\lim\limits_{y\to 0}0=0$，$\lim\limits_{x\to 0}\lim\limits_{y\to 0}\frac{2xy}{x^2+y^2}=\lim\limits_{x\to 0}0=0.$

先求 x→0 时

P.67 例 8 $f(x,y)=x\sin\frac{1}{y}+y\sin\frac{1}{x}$，求累次极限 $\lim\limits_{x\to 0}\lim\limits_{y\to 0}f(x,y)$、$\lim\limits_{y\to 0}\lim\limits_{x\to 0}f(x,y)$ 与二重极限

$\lim\limits_{\substack{x\to 0\\ y\to 0}} f(x,y)$。

解：$\lim\limits_{y\to 0}\sin\dfrac{1}{y}$不存在 $\Rightarrow \lim\limits_{y\to 0}\left(x\sin\dfrac{1}{y}+y\sin\dfrac{1}{x}\right)$不存在 $\Rightarrow \lim\limits_{x\to 0}\lim\limits_{y\to 0}\left(x\sin\dfrac{1}{y}+y\sin\dfrac{1}{x}\right)$不存在.

$\lim\limits_{x\to 0}\sin\dfrac{1}{x}$不存在 $\Rightarrow \lim\limits_{x\to 0}\left(x\sin\dfrac{1}{y}+y\sin\dfrac{1}{x}\right)$不存在 $\Rightarrow \lim\limits_{y\to 0}\lim\limits_{x\to 0}\left(x\sin\dfrac{1}{y}+y\sin\dfrac{1}{x}\right)$不存在

但 $0\le|f(x,y)|\le|x|+|y|$, 由 $\lim\limits_{\substack{x\to 0\\ y\to 0}}(|x|+|y|)=0$, 则 $\lim\limits_{\substack{x\to 0\\ y\to 0}} f(x,y)=0$.

由例 7 知，两个累次极限存在，但二重极限不存在.

由例 8 知，二重极限存在，但两个累次极限不存在，但有如下关系：

前提都存在.

P.67 定理 8.1 若累次极限 $\lim\limits_{x\to x_0}\lim\limits_{y\to y_0} f(x,y)$, $\lim\limits_{y\to y_0}\lim\limits_{x\to x_0} f(x,y)$和二重极限 $\lim\limits_{\substack{x\to x_0\\ y\to y_0}} f(x,y)$都存在，则

三者相等（证明略）.

推论 若 $\lim\limits_{x\to x_0}\lim\limits_{y\to y_0} f(x,y)$ ，$\lim\limits_{y\to y_0}\lim\limits_{x\to x_0} f(x,y)$ 存在但不相等，则 $\lim\limits_{\substack{x\to x_0\\ y\to y_0}} f(x,y)$ 不存在。

$$\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}} \Delta z = 0$$

x偏增量 $\Delta_x z = f(x,y_0) - f(x_0,y_0) = f(x_0+\Delta x, y_0) - f(x_0,y_0)$

y偏增量 $\Delta_y z = f(x_0,y) - f(x_0,y_0) = f(x_0,y_0+\Delta y) - f(x_0,y_0)$

求二元（或多元）函数极限方法

1. 不等式 + 夹逼
2. 变量替换化为已知极限或一元函数极限，极坐标示
3. 初等函数连续性，极限的四则运算.
4. 初等变形，指数 → 对数
5. 能看出极值 → $\varepsilon-\delta$ 语言

二元(多元)函数极限不存在

1. 任向路径极限与幅角(斜率)有关
2. 证明某特殊极限不存在
3. 两极限存在但不相等
4. 空心邻域内存在,累次极限不相等

不同路径+归结原则

偏导数与全微分

§2.1 偏导数

$$\lim_{\Delta x\to 0}\frac{\Delta_x z}{\Delta x}=\lim_{\Delta x\to 0}\frac{f(x_0+\Delta x,y_0)-f(x_0,y_0)}{\Delta x}=\lim_{x\to x_0}\frac{f(x,y_0)-f(x_0,y_0)}{x-x_0}$$

$z=f(x,y)$ 在 $P(x_0,y_0)$ 某邻域有定义

$z=f(x,y)$ $f(x,y_0)=g(x)$ 一元函数

$\frac{d}{dx}f(x,y_0)$ → 一元函数

$\frac{\partial}{\partial x}f(x,y)$ → 2或多元函数

区别:几个自变量

求导 对谁自变量偏

↓

$P(x_0,y_0)$

记作 $f_x'(x_0,y_0)$ 或 $\left.\frac{\partial z}{\partial x}\right|_{\substack{x=x_0\\y=y_0}}$ 或 $\left.z_x'\right|_{\substack{x=x_0\\y=y_0}}$ 或 $\left.\frac{\partial}{\partial x}f(x,y)\right|_{\substack{x=x_0\\y=y_0}}$

$$f_x'(x_0,y_0)=\left.\frac{d}{dx}f(x,y_0)\right|_{x=x_0}$$

同理定义 $z=f(x,y)$ 在 $P_0(x_0,y_0)$ 对 y 偏导

二、计算偏导数

(1) 求定义

(2) 求 $\frac{d}{dx}f(x,y_0)$ 把 $x=x_0$ 代入

(3) 求 $f_x'(x,y)$ 再带入 $x=x_0$ $y=y_0$

先部分代入再求导

先求导再代入

P.71 例 1 $f(x,y)=x+(y-1)\arcsin\sqrt{\frac{x}{y}}$，求 $f'_x\left(\frac{1}{2},1\right), f'_y\left(\frac{1}{2},1\right)$

解法一　用偏导数定义

$$f'_x\left(\frac{1}{2},1\right)=\lim_{x\to\frac{1}{2}}\frac{f(x,1)-f\left(\frac{1}{2},1\right)}{x-\frac{1}{2}}=\lim_{x\to\frac{1}{2}}\frac{x-\frac{1}{2}}{x-\frac{1}{2}}=1.$$

$$f'_y\left(\frac{1}{2},1\right)=\lim_{y\to1}\frac{f\left(\frac{1}{2},y\right)-f\left(\frac{1}{2},1\right)}{y-1}=\lim_{y\to1}\frac{\frac{1}{2}+(y-1)\arcsin\sqrt{\frac{1}{2y}}-\frac{1}{2}}{y-1}=\lim_{y\to1}\arcsin\sqrt{\frac{1}{2y}}=\frac{\pi}{4}$$

解法二　求导函数 $\frac{d}{dx}f(x,y_0)$，然后把 $x=x_0$ 代入；

$$f'_x\left(\frac{1}{2},1\right)=\frac{d}{dx}f(x,1)\Big|_{x=\frac{1}{2}}=x'\big|_{x=\frac{1}{2}}=1,$$

$$f'_y\left(\frac{1}{2},1\right)=\frac{d}{dy}f\left(\frac{1}{2},y\right)\Big|_{y=1}=\left[\frac{1}{2}+(y-1)\arcsin\sqrt{\frac{1}{2y}}\right]'\Big|_{y=1}=\frac{\pi}{4}.$$

解法三　先求出偏导函数

$$f'_x(x,y)=1+\frac{1}{2}(y-1)\frac{1}{\sqrt{1-\frac{x}{y}}\sqrt{\frac{x}{y}}}\cdot\frac{1}{y}\Rightarrow f'_x\left(\frac{1}{2},1\right)=1$$

$$f'_y(x,y)=\arcsin\sqrt{\frac{x}{y}}+\frac{1}{2}(y-1)\frac{1}{\sqrt{1-\frac{x}{y}}}\cdot\frac{1}{\sqrt{\frac{x}{y}}}\cdot\frac{(-x)}{y^2}\Rightarrow f'_y\left(\frac{1}{2},1\right)=\arcsin\frac{\sqrt{2}}{2}=\frac{\pi}{4}$$

P.71 例 2　$u=x^y$，求 $\frac{\partial u}{\partial x},\frac{\partial u}{\partial y}$.

解　把 y 看成常数，这时 u 是 x 的幂函数，对 x 求偏导 $\frac{\partial u}{\partial x}=yx^{y-1}$.

把 x 看成常数，这时 u 是 y 的指数函数，对 y 求偏导 $\frac{\partial u}{\partial x}=x^y\ln x$.

P.72 例 4　$u=x^{y^z}$　求 $\frac{\partial u}{\partial x},\frac{\partial u}{\partial y},\frac{\partial u}{\partial z},\frac{\partial u}{\partial z}\Big|_{(1,1,1)}$

解　把 y,z 看成常数，对 x 求偏，有 $\frac{\partial u}{\partial x}=y^zx^{y^z-1}=\frac{u}{x}y^z$

把 x,z 看成常数，有 $\frac{\partial u}{\partial y}=zx^{y^z}\ln x\cdot y^{z-1}=zuy^{z-1}\ln x.$

把 x,y 看成常数，有 $\frac{\partial u}{\partial z}=x^{y^z}\ln x\cdot y^z\ln y=uy^z\ln x\cdot\ln y$，　$\frac{\partial u}{\partial z}\Big|_{(1,1,1)}=0.$

P.72 例 3　$u=(x^2+y^2)^{xy}$

解　$u=e^{xy\ln(x^2+y^2)}$

$$\Rightarrow\frac{\partial u}{\partial y}=e^{xy\ln(x^2+y^2)}y\left[\ln(x^2+y^2)+x\cdot\frac{2x}{x^2+y^2}\right]=(x^2+y^2)^{xy}y\left[\ln(x^2+y^2)+\frac{2x^2}{x^2+y^2}\right]$$

由 x,y 地位一样，利用轮换，把 x 换成 y，y 换成 x，有

$$\frac{\partial u}{\partial y}=(x^2+y^2)^{xy}x\left[\ln(x^2+y^2)+\frac{2y^2}{x^2+y^2}\right]$$

P.72 例 5　设 $z=yf(x^2-y^2)$，其中 f 是导数存在的任意一元函数，

证明函数 z 满足方程 $\frac{1}{x}\cdot\frac{\partial z}{\partial x}+\frac{1}{y}\cdot\frac{\partial z}{\partial y}=\frac{z}{y^2}$.

证：

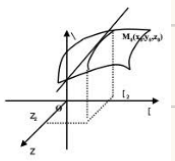
$$\frac{\partial z}{\partial x}=\frac{\partial}{\partial x}[yf(x^2-y^2)]=y\frac{\partial}{\partial x}f(x^2-y^2)=yf'(x^2-y^2)\cdot2x,$$

$$\frac{\partial z}{\partial y}=\frac{\partial}{\partial y}[yf(x^2-y^2)]=f(x^2-y^2)+yf'(x^2-y^2)\cdot(-2y),$$

$$\Rightarrow\frac{1}{x}\cdot\frac{\partial z}{\partial x}+\frac{1}{y}\cdot\frac{\partial z}{\partial y}=2yf'(x^2-y^2)+\frac{1}{y}f(x^2-y^2)-2yf'(x^2-y^2)=\frac{z}{y^2}$$

三　偏导数的几何意义

曲面 $z=f(x,y)$，设 $M_0(x_0,y_0,z_0)$ 是曲面上一点



$f_x'(x_0,y_0)=\frac{d}{dx}f(x,y_0)\Big|_{x=x_0}\Rightarrow$ 一元函数 $f(x,y_0)$ 在 $x=x_0$ 的导数

$\begin{cases}z=f(x,y)\\y=y_0\end{cases}\Rightarrow$ 一元函数 $f(x,y_0)$

$\Rightarrow$偏导数 $f_x'(x_0,y_0)$ 的几何意义是表示 $\begin{cases}z=f(x,y)\\y=y_0\end{cases}$ 在 M_0 处的切线对 Ox 轴的斜率。

同理：

偏导数 $f_y'(x_0,y_0)$ 的几何意义是表示 $\begin{cases}z=f(x,y)\\x=x_0\end{cases}$ 在 M_0 处的切线对 Oy 轴的斜率。

P.73 例 6 讨论函数 $f(x,y)=\begin{cases}\frac{2xy}{x^2+y^2}, & x^2+y^2\neq0\\0, & x^2+y^2=0\end{cases}$ 在(0,0)点

偏导数是否存在，函数 f 是否连续？

分析　按定义求函数 f 在(0,0)处的偏导数：$f_x'(0,0)=0;f_y'(0,0)=0\Rightarrow$函数 f 在点(0,0)处的偏导数都存在，但由上节 P.66 例 7 知，此函数 f 在(0,0)处极限不存在，所以函数 f 在(0,0)处不连续.

注：函数 $\sqrt{x^2+y^2}$ 在(0,0)处连续，但偏导数都不存在.

结论：二元（多元）函数：可导　不一定 连续

四、高阶偏导数

与一元函数的高阶导数一样，可以定义多元函数的高阶偏导数.

若函数 $z=f(x,y)$ 的偏导数 $\frac{\partial z}{\partial x},\frac{\partial z}{\partial y}$ 存在，称为一阶偏导数，它们

仍是 x,y 的函数，如果它们对 x,y 的偏导仍存在，得到

$$\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right),\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right),\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right),\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right),$$ 称之为 $z=f(x,y)$ 的二阶

偏导数，记作

$$z''_{xx}=\frac{\partial^2z}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)=f''_{xx}(x,y)\qquad z''_{xy}=\frac{\partial^2z}{\partial x\partial y}=\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)=f''_{xy}(x,y)$$

$$z''_{yx}=\frac{\partial^2z}{\partial y\partial x}=\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)=f''_{yx}(x,y)\qquad z''_{yy}=\frac{\partial^2z}{\partial y^2}=\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right)=f''_{yy}(x,y)$$

其中 $f''_{xy}(x,y)$ 和 $f''_{yx}(x,y)$ 称为二阶混合偏导数. 类似可定义三阶及三阶以上的偏导数，如

$$\frac{\partial}{\partial x}\left(\frac{\partial^2z}{\partial x^2}\right)=\frac{\partial^3z}{\partial x^3}=f'''_{xxx}(x,y);\ \frac{\partial}{\partial y}\left(\frac{\partial^2z}{\partial x^2}\right)=\frac{\partial^3z}{\partial x^2\partial y}=f'''_{xxy}(x,y);$$

二阶及二阶以上的偏导数，统称为高阶偏导数.

注意：今天做作业之前请先阅读 P.74 例 7 和例 8。

P.74 例 7　$z=\arctan\frac{y}{x}$，求所有的一阶及二阶偏导数.

提示：$\frac{\partial z}{\partial x}=-\frac{y}{x^2+y^2},\frac{\partial z}{\partial y}=\frac{x}{x^2+y^2},$

$$\frac{\partial^2z}{\partial x^2}=\frac{\partial}{\partial x}\left(-\frac{y}{x^2+y^2}\right)=\frac{2xy}{(x^2+y^2)^2},\qquad\frac{\partial^2z}{\partial y^2}=\frac{\partial}{\partial y}\left(\frac{x}{x^2+y^2}\right)=-\frac{2xy}{(x^2+y^2)^2},$$

$$\frac{\partial^2z}{\partial x\partial y}=\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial}{\partial y}\left(-\frac{y}{x^2+y^2}\right)=\frac{y^2-x^2}{(x^2+y^2)^2},$$

$$\frac{\partial^2z}{\partial y\partial x}=\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)=\frac{\partial}{\partial x}\left(\frac{x}{x^2+y^2}\right)=\frac{y^2-x^2}{(x^2+y^2)^2},$$

P.74 例 8　$z=x^3+x^2y-y^2$，求 $\frac{\partial^2z}{\partial x\partial y},\frac{\partial^2z}{\partial y\partial x}$.

解　$\frac{\partial z}{\partial x}=3x^2+2xy;\ \frac{\partial^2z}{\partial x\partial y}=2x;$

$\frac{\partial z}{\partial y}=x^2-2y;\ \frac{\partial^2z}{\partial y\partial x}=2x.$

例: $f(x,y)=f(x)=\begin{cases}\dfrac{xy(x^2-y^2)}{x^2+y^2}, & \text{当 } (x,y)\neq(0,0)\\ 0, & \text{当 } (x,y)=(0,0)\end{cases}$

求 $f''_{yx}(0,0)$, $f''_{xy}(0,0)$

$$\lim_{x\to 0}\frac{f(x,y)-f(0,y)}{x}=\lim_{x\to 0}\frac{xy\frac{x^2-y^2}{x^2+y^2}-0}{x}=-y=f'_x(0,y)$$

$$f''_{xy}(0,0)=\frac{d}{dy}[f'_x(0,y)]_{y=0}=-1$$

同理 $f''_{yx}(0,0)=\frac{d}{dx}[f'_y(x,0)]_{x=0}=1$

$$f''_{xy}(0,0)\neq f''_{yx}(0,0)$$

例: 不连续 → 二阶偏导不相等

P.74 定理 8.3 若函数$z=f(x,y)$的二阶偏导（函数）$f''_{xy}(x,y),f''_{yx}(x,y)$都在$(x_0,y_0)$处连续，则 $f''_{xy}(x_0,y_0)=f''_{yx}(x_0,y_0)$.

提示： 由$f''_{xy}(x,y),f''_{yx}(x,y)$都在$(x_0,y_0)$处连续，则$f''_{xy}(x,y),f''_{yx}(x,y)$在$P_0(x_0,y_0)$的某邻域$U(P_0)$内有定义，任给$\Delta x\neq 0,\Delta y\neq 0$，并取$|\Delta x|$，$|\Delta y|$充分小，使$(x_0+\Delta x,y_0),(x_0,y_0+\Delta y),(x_0+\Delta x,y_0+\Delta y)\in U(P_0)$，

令 $\varphi(x)=f(x,y_0+\Delta y)-f(x,y_0)$，

$\psi(y)=f(x_0+\Delta x,y)-f(x_0,y)$，

令 $W=f(x_0+\Delta x,y_0+\Delta y)+f(x_0,y_0)-f(x_0+\Delta x,y_0)-f(x_0,y_0+\Delta y)$

注：W有两种不同组合形式计，对这两种不同组合形式分别讨论：

(1) $W=[f(x_0+\Delta x,y_0+\Delta y)-f(x_0+\Delta x,y_0)]-[f(x_0,y_0+\Delta y)-f(x_0,y_0)]$

$=\varphi(x_0+\Delta x)-\varphi(x_0)=\varphi'(x_0+\theta_1\Delta x)\Delta x$

$=[f'_x(x_0+\theta_1\Delta x,y_0+\Delta y)-f'_x(x_0+\theta_1\Delta x,y_0)]\Delta x(0<\theta_1<1)$

$=f''_{xy}(x_0+\theta_1\Delta x,y_0+\theta_2\Delta y)\Delta y\Delta x\ (0<\theta_2<1)$.

同理(2) $W=[f(x_0+\Delta x,y_0+\Delta y)-f(x_0,y_0+\Delta y)]-[f(x_0+\Delta x,y_0)-f(x_0,y_0)]$

$=\psi(y_0+\Delta y)-\psi(y_0)=\psi'(y_0+\theta_3\Delta y)\Delta y$

$=[f'_y(x_0+\Delta x,y_0+\theta_3\Delta y)-f'_y(x_0,y_0+\theta_3\Delta y)]\Delta y$

$=f''_{yx}(x_0+\theta_4\Delta x,y_0+\theta_3\Delta y)\Delta x\Delta y\ (0<\theta_3,\theta_4<1)$

则得： $f''_{xy}(x_0+\theta_1\Delta x,y_0+\theta_2\Delta y)\Delta y\Delta x=f''_{yx}(x_0+\theta_4\Delta x,y_0+\theta_3\Delta y)\Delta x\Delta y$

$\because \Delta x\Delta y\neq 0$

$\therefore f''_{xy}(x_0+\theta_1\Delta x,y_0+\theta_2\Delta y)=f''_{yx}(x_0+\theta_4\Delta x,y_0+\theta_3\Delta y)\ (0<\theta_1,\theta_2,\theta_3,\theta_4<1)$

$\because f''_{xy}(x,y),f''_{yx}(x,y)$都在$P_0(x_0,y_0)$处连续 $\therefore f''_{xy}(x_0,y_0)=f''_{yx}(x_0,y_0)$

注：这个定理的结论对n元函数的混合偏导数也成立：对n元函数在一点存在直到m阶的连续偏导数，则在这一点上$k(\leq m)$阶混合偏导数与求得顺序无关.

分析补充题二阶混合偏导不相等的原因：

* 补充 设函数 $f(x,y)=f(x)=\begin{cases}\dfrac{xy(x^2-y^2)}{x^2+y^2}, & \text{当}(x,y)\neq(0,0)\\ 0, & \text{当}(x,y)=(0,0)\end{cases}$

$$f'_x(x,y)=\begin{cases}\dfrac{x^4y+4x^2y^3-y^5}{(x^2+y^2)^2}, & \text{当}(x,y)\neq(0,0)\\ 0, & \text{当}(x,y)=(0,0)\end{cases}$$

$$f'_y(x,y)=\begin{cases}\dfrac{x^5-4x^3y^2-xy^4}{(x^2+y^2)^2}, & \text{当}(x,y)\neq(0,0)\\ 0, & \text{当}(x,y)=(0,0)\end{cases}$$

$$f''_{xy}(x,y)=\begin{cases}\dfrac{x^6+9x^2y^2(x^2-y^2)-y^6}{(x^2+y^2)^3}, & \text{当}(x,y)\neq(0,0)\\ -1, & \text{当}(x,y)=(0,0)\end{cases}$$

$$f''_{yx}(x,y)=\begin{cases}\dfrac{x^6+9x^2y^2(x^2-y^2)-y^6}{(x^2+y^2)^3}, & \text{当}(x,y)\neq(0,0)\\ 1, & \text{当}(x,y)=(0,0)\end{cases}$$

$$\lim_{\substack{y=kx\\x\to 0}}f''_{xy}(x,y)=\lim_{\substack{y=kx\\x\to 0}}f''_{yx}(x,y)=\lim_{\substack{y=kx\\x\to 0}}\frac{x^6+9x^2y^2(x^2-y^2)-y^6}{(x^2+y^2)^3}$$

$$=\lim_{\substack{y=kx\\x\to 0}}\frac{x^6+9k^2(1-k^2)x^6-k^6y^6}{(1+k^2)^3x^6}=\frac{1+9k^2(1-k^2)-k^6}{(1+k^2)^3}$$

$\therefore \lim_{(x,y)\to(0,0)}f''_{xy}(x,y)$, $\lim_{(x,y)\to(0,0)}f''_{yx}(x,y)$ 不存在.

$\therefore f''_{xy}(x,y),f''_{yx}(x,y)$都在$(0,0)$处不连续.

全微分四则运算

(i) $d(u\pm v)=du\pm dv$

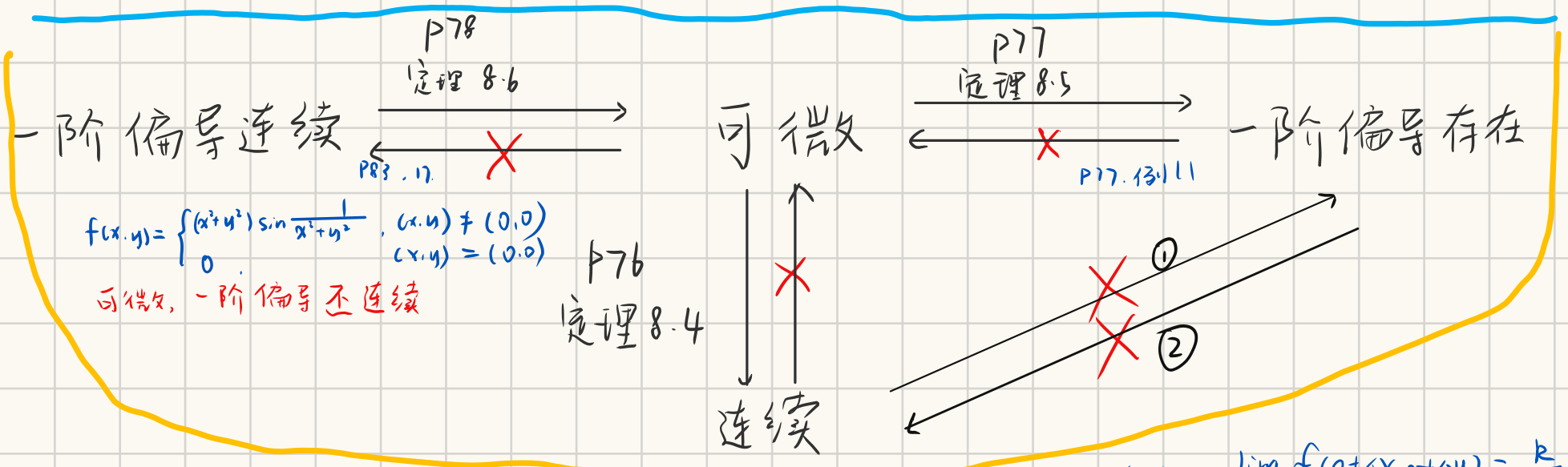
(ii) $d(uv)=vdu+udv$

(iii) $d\left(\frac{u}{v}\right)=\frac{vdu-udv}{v^2}\ (v\neq 0)$

设$u=u(x,y),v=v(x,y)$,由条件知，uv可微，且

$$d(uv)=\frac{\partial}{\partial x}(uv)dx+\frac{\partial}{\partial y}(uv)dy=\left(\frac{\partial u}{\partial x}v+u\frac{\partial v}{\partial x}\right)dx+\left(\frac{\partial u}{\partial y}v+u\frac{\partial v}{\partial y}\right)dy$$

$$=\left(\frac{\partial u}{\partial x}dx+\frac{\partial u}{\partial y}dy\right)v+\left(\frac{\partial v}{\partial x}dx+\frac{\partial v}{\partial y}dy\right)u=vdu+udv$$



$f(x,y)=\begin{cases}(x^2+y^2)\sin\frac{1}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$

可微，一阶偏导不连续

① $f(x,y)=\sqrt{x^2+y^2}$

$\lim\limits_{(x,y)\to(x_0,y_0)} f(x,y)=\sqrt{x_0^2+y_0^2}=f(x_0,y_0)$ 连续

$f_x'(0,0)=\lim\limits_{\Delta x\to 0}\frac{|\Delta x|}{x}=\begin{cases}1, & x\to 0^+\\ -1, & x\to 0^-\end{cases}$ 不存在

② $f(x,y)=\begin{cases}\frac{xy}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$

0处值极限：$\lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to k\Delta x}} f(0+\Delta x,0+\Delta y)=\frac{k}{1+k^2}$ 与k有关

$f_x'=\frac{f(0+\Delta x,0)-f(0,0)}{\Delta x}=\frac{\frac{\Delta x\cdot 0}{(\Delta x)^2+0}-0}{\Delta x}$

同理 $f_y'=0$ $(\Delta x\neq 0)$ 一阶导数存在 $=0$

(P67) **连续**：$f(P)=f(x,y)$ 在某邻域或 $U(P_0)$ 有意义

① $\lim\limits_{\substack{x\to x_0\\ y\to y_0}} f(x,y)=f(x_0,y_0)$ 则称 $f(P)$ 在 $P(x_0,y_0)$ 处连续

$\Delta z=f(x,y)-f(x_0,y_0)=f(x_0+\Delta x,y_0+\Delta y)-f(x_0,y_0)$ 全增量

或 ② $\lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}} \Delta z=0$

(P76) **可微**：全增量 $\Delta z=f(x+\Delta x,y+\Delta y)-f(x,y)$ 可表示为

$\Delta z=A\Delta x+B\Delta y+o(\rho)$ $(\rho=\sqrt{(\Delta x)^2+(\Delta y)^2}\to 0)$

A B 与 $\Delta x,\Delta y$ 无关，又与 (x,y) 有关，则称 $f(x,y)$ 在 (x,y) 有关.

Δz 线性部 全微分 $dz=A\Delta x+B\Delta y$

是否可微

即 $\lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\frac{\Delta z-(A\Delta x+B\Delta y)}{\rho}$ 或 $\frac{\Delta z-dz}{\rho}$ 或 $\frac{\Delta z-f_x'(x,y)\Delta x-f_y'(x,y)\Delta y}{\sqrt{(\Delta x)^2+(\Delta y)^2}}$ 是否等于0

(P69) **一阶偏导存在**

$\lim\limits_{\Delta x\to 0}\frac{\Delta_x z}{\Delta x}=\lim\limits_{\Delta x\to 0}\frac{f(x_0+\Delta x,y_0)-f(x_0,y_0)}{\Delta x}=\lim\limits_{x\to x_0}\frac{f(x,y_0)-f(x_0,y_0)}{x-x_0}$ 存在

(P78) **一阶偏导连续**

$f_x'(x_0+\Delta x,y_0+\Delta y)-f_x'(x_0,y_0)=\varepsilon_1$ $\lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\varepsilon_1=0$

证明反例见下

定理8.6 一阶偏导连续 ⟶ 可微

定理 8.6（可微的充分条件）若函数$z=f(x,y)$的偏导数$f_x'(x,y), f_y'(x,y)$在点(x_0,y_0)处连续，则函数$z=f(x,y)$在点(x_0,y_0)处可微.

提示：
$$\begin{aligned}\Delta z &= f(x_0+\Delta x, y_0+\Delta y)-f(x_0,y_0)\\ &= [f(x_0+\Delta x, y_0+\Delta y)-f(x_0, y_0+\Delta y)]+[f(x_0, y_0+\Delta y)-f(x_0,y_0)]\\ &= f_x'(x_0+\theta_1\Delta x, y_0+\Delta y)\Delta x + f_y'(x_0, y_0+\theta_2\Delta y)\Delta y \quad (0<\theta_1,\theta_2<1)\end{aligned}$$

由 P. 66 (8.1)式得：$f_x'(x_0+\theta_1\Delta x, y_0+\Delta y)=f_x'(x_0,y_0)+\varepsilon_1$ $\left(\text{其中}\lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\varepsilon_1=0\right)$

同理 $f_y'(x_0,y_0+\theta_2\Delta y)=f_y'(x_0,y_0)+\varepsilon_2$ $\left(\text{其中}\lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\varepsilon_2=0\right)$，则

$$\Delta z = f_x'(x_0,y_0)\Delta x + f_y'(x_0,y_0)\Delta y + \varepsilon_1\Delta x+\varepsilon_2\Delta y,$$

$$\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\frac{\varepsilon_1\Delta x+\varepsilon_2\Delta y}{\rho}=\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\left(\frac{\varepsilon_1\Delta x}{\sqrt{\Delta x^2+\Delta y^2}}+\frac{\varepsilon_2\Delta y}{\sqrt{\Delta x^2+\Delta y^2}}\right)=0,\text{得}$$

$\Delta z = f_x'(x_0,y_0)\Delta x + f_y'(x_0,y_0)\Delta y + o(\rho)\ (\rho\to 0)$.因此，$z=f(x,y)$在点$P_0(x_0,y_0)$处可微.

注：$o(\rho)$也可写成：$o(\rho)=\varepsilon_1\Delta x+\varepsilon_2\Delta y$ $\left(\text{其中}\lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\varepsilon_1=0, \lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\varepsilon_2=0\right)$

$$\because o(\rho)=\frac{o(\rho)\rho^2}{\rho^2}=\frac{o(\rho)}{\rho}\cdot\frac{\Delta x}{\rho}\cdot\Delta x+\frac{o(\rho)}{\rho}\cdot\frac{\Delta y}{\rho}\cdot\Delta y,$$

设$\varepsilon_1\triangleq\frac{o(\rho)}{\rho}\cdot\frac{\Delta x}{\rho}, \varepsilon_2\triangleq\frac{o(\rho)}{\rho}\cdot\frac{\Delta y}{\rho}$,

则：Δz也可写成：

$$\Delta z = f_x'(x_0,y_0)\Delta x + f_y'(x_0,y_0)\Delta y+\varepsilon_1\Delta x+\varepsilon_2\Delta y$$

其中$\lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\varepsilon_1=0, \lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\varepsilon_2=0$.这个等式我们又称为全增量公式.

判别可微新方法：一个多元函数$z=f(x,y)$在点(x_0,y_0)处是否可微，只要求出偏导函数$f_x'(x,y), f_y'(x,y)$，若$f_x'(x,y), f_y'(x,y)$在点(x_0,y_0)处连续，则$z=f(x,y)$在点(x_0,y_0)处可微.

P83. 17. 可微 ⟶ 一阶偏导连续

P. 83 17 证明$f(x,y)=\begin{cases}(x^2+y^2)\sin\frac{1}{x^2+y^2}, & \text{当}(x,y)\neq(0,0)\\ 0, & \text{当}(x,y)=(0,0)\end{cases}$在点(0,0)处可微。

提示：(1)考虑$f_x'(x,y), f_y'(x,y)$在点(0,0)的性:

$$f_x'(x,y)=\begin{cases}2x\sin\frac{1}{x^2+y^2}-\frac{2x}{x^2+y^2}\cos\frac{1}{x^2+y^2}, & \text{当}(x,y)\neq(0,0)\\ 0, & \text{当}(x,y)=(0,0)\end{cases}$$

$$f_y'(x,y)=\begin{cases}2y\sin\frac{1}{x^2+y^2}-\frac{2y}{x^2+y^2}\cos\frac{1}{x^2+y^2}, & \text{当}(x,y)\neq(0,0)\\ 0, & \text{当}(x,y)=(0,0)\end{cases}$$

当$(x,y)\neq(0,0)$时，令$x=r\cos\theta, y=r\sin\theta$, $f_x'(x,y)=2r\cos\theta\sin\frac{1}{r^2}-\frac{2r\cos\theta}{r^2}\cos\frac{1}{r^2}$

$$r\to 0\Rightarrow\begin{cases}2r\cos\theta\sin\frac{1}{r^2}\to 0\\ \frac{2r\cos\theta}{r^2}\cos\frac{1}{r^2}\text{无界}\end{cases}\Rightarrow\lim_{\substack{x\to 0\\ y\to 0}}f_x'(x,y)\text{不存在}\Rightarrow f_x'(x,y)\text{在}(0,0)\text{处不连续}$$

同理$f_y'(x,y)$在点(0,0)处不连续

(2)考虑$f(x,y)$在点(0,0)的可微性. $\left(\rho\triangleq\sqrt{\Delta x^2+\Delta y^2}\right)$

$$dz=f_x'(0,0)\Delta x+f_y'(0,0)\Delta y=0\cdot\Delta x+0\cdot\Delta y=0,$$

$$\Delta z=f(0+\Delta x,0+\Delta y)-f(0,0)=[\Delta x^2+\Delta y^2]\sin\frac{1}{\Delta x^2+\Delta y^2}-0=\rho^2\sin\frac{1}{\rho},$$

$$\Rightarrow\frac{\Delta z-A\Delta x-B\Delta y}{\rho}=\frac{\rho^2\sin\frac{1}{\rho}}{\rho}=\rho\sin\frac{1}{\rho}\to 0(\rho\to 0).\Leftrightarrow\Delta z=0\cdot\Delta x+0\cdot\Delta y+o(\rho)(\rho\to 0)$$

$\Rightarrow f(x,y)$在点(0,0)处可微。

注：本题函数在点(0,0)的邻域中有偏导函数$f_x'(x,y)$和$f_y'(x,y)$在点(0,0)处是不连续的且在此点的任何邻域中是无界的；然而此函数在点(0,0)处可微。

思考题：$f(x,y)=\begin{cases}(x^2+y^2)\sin\frac{1}{\sqrt{x^2+y^2}}, & \text{当}(x,y)\neq(0,0)\\ 0, & \text{当}(x,y)=(0,0)\end{cases}$ 在点(0,0)处可微，

偏导数在(0,0)处不连续。

P77 定理8.5

可微

⟶

偏导数存在 ☆

定理 8.5(可微的必要条件) 若$z=f(x,y)$在点(x,y)处可微，则$z=f(x,y)$在点(x,y)的两个偏导数$f_x'(x,y), f_y'(x,y)$都存在，且$A=f_x'(x,y), B=f_y'(x,y)$.

分析：可微 $\Rightarrow\Delta z=A\Delta x+B\Delta y+o(\rho)\ (\rho\to 0)$

$$\Delta y=0\Rightarrow\Delta_x z=A\Delta x+o(|\Delta x|)\Rightarrow\lim_{\Delta x\to 0}\frac{\Delta_x z}{\Delta x}=A=f_x'(x,y).$$

$$\Delta x=0\Rightarrow\Delta_y z=B\Delta y+o(|\Delta y|)\Rightarrow\lim_{\Delta y\to 0}\frac{\Delta_y z}{\Delta y}=B=f_y'(x,y).$$

$$\therefore dz=f_x'(x,y)\Delta x+f_y'(x,y)\Delta y$$

$$\begin{cases}z\equiv x\Rightarrow dx=1\cdot\Delta x+0\cdot\Delta y\\ z\equiv y\Rightarrow dy=0\cdot\Delta x+1\cdot\Delta y\end{cases}\Rightarrow dz=f_x'(x,y)dx+f_y'(x,y)dy$$

该定理的逆命题不一定成立.

P77 例11 偏导数存在 ⟶ 可微

P.77 例 11 研究$f(x,y)=\begin{cases}\dfrac{2xy}{\sqrt{x^2+y^2}}, & x^2+y^2\neq 0\\ 0, & x^2+y^2=0\end{cases}$在原点的可微性.

提示: $f_x'(0,0)=\lim\limits_{x\to 0}\dfrac{f(x,0)-f(0,0)}{x}=0$ 同理$f_y'(0,0)=0$.

$$\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\frac{\Delta z-dz}{\rho}=\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\frac{\dfrac{2\Delta x\Delta y}{\sqrt{\Delta x^2+\Delta y^2}}-(0\cdot\Delta x+0\cdot\Delta y)}{\sqrt{\Delta x^2+\Delta y^2}}=\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\frac{2\Delta x\Delta y}{\Delta x^2+\Delta y^2}$$

由本章第二节 P.66 例 7 知极限不存在，因此函数f在原点不可微. 这个例子(定理 8.4、定理 8.5 的反例)说明，偏导数即使存在、即使是连续，函数也不一定可微. 验证多元函数不可微有下述方法:

(i)若$f(x,y)$在点(x,y)处不连续，则f在点(x,y)处不可微;

(ii)若$f(x,y)$在点(x,y)处至少有一个偏导数不存在，则f在点(x,y)处不可微;

(iii)若$f(x,y)$在点(x,y)处两个偏导数都存在，而

$\lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\dfrac{\Delta z-dz}{\rho}$极限不存在或极限存在但不为零，则$f$在点$(x,y)$处不可微.

P76 定理 8.4 可微 ⟶ 连续

二、可微的必要条件、充分条件

定理 8.4 若$z=f(x,y)$在点(x,y)处可微，则$z=f(x,y)$在点(x,y)处连续，反之不成立.

分析: 可微 $\Rightarrow \Delta z=A\Delta x+B\Delta y+o(\rho)\ (\rho\to 0)\Rightarrow \lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\Delta z=0$

由一元函数中连续不一定可微⇒在多元函数中连续也不一定可微.

连续 ⟶ 可微

$$f(x,y)=\begin{cases}\dfrac{xy}{\sqrt{x^2+y^2}}, & (x,y\neq 0)\\ 0, & (x,y)=(0,0)\end{cases}$$

不可微: $f_x'(0,0)=\lim\limits_{\Delta x\to 0}\dfrac{f(0+\Delta x,0)-f(0,0)}{\Delta x-0}=\dfrac{0-0}{\Delta x}=0$ 注意点顶0

同理 $f_y'(0,0)=0$

$$\Delta z=f(0+\Delta x,0+\Delta y)-f(0,0)=\frac{\Delta x\Delta y}{\sqrt{(\Delta x)^2+(\Delta y)^2}}$$

$$\frac{\Delta z-0\cdot\Delta x-0\cdot\Delta y}{\sqrt{(\Delta x)^2+(\Delta y)^2}}=\lim_{\substack{\Delta x\to 0\\ \Delta y\to k\Delta x}}\frac{\Delta x\Delta y}{(\Delta x)^2+(\Delta y)^2}$$ 非定值

∴ 不可微

连续. $x=r\cos\theta$, $y=r\sin\theta$

$$\lim_{(x,y)\to 0}f(x,y)=\lim_{r\to 0}\frac{r\cos\theta\, r\sin\theta}{r}=\lim_{r\to 0}r\cos\theta\sin\theta=0$$

$\therefore \lim\limits_{(x,y)}f(x,y)=f(0,0)=0\Rightarrow$ 连续

四、全微分在近似计算和误差估计中的应用

由全微分的定义，当$|\Delta x|,|\Delta y|$很小时，有 $\Delta z \approx dz = f_x'(x,y)\Delta x + f_y'(x,y)\Delta y.$

得：$f(x+\Delta x, y+\Delta y) - f(x,y) \approx f_x'(x,y)\Delta x + f_y'(x,y)\Delta y.$

设函数$z=f(x,y)$，若测得x的近似值为x_0, y的近似值为y_0，测量的近似值x_0, y_0分别代替x, y来计算函数值z，设$|x-x_0| \le \delta_1, |y-y_0| \le \delta_2$,就会引起绝对误差

$$|\Delta z| = |f(x,y) - f(x_0,y_0)| \approx |dz| = \left|f_x'(x_0,y_0)\Delta x + f_y'(x_0,y_0)\Delta y\right|$$

$$\le |f_x'(x_0,y_0)||\Delta x| + \left|f_y'(x_0,y_0)\right||\Delta y| \le |f_x'(x_0,y_0)|\delta_1 + \left|f_y'(x_0,y_0)\right|\delta_2;$$

相对误差：

$$\left|\frac{\Delta z}{z_0}\right| \approx \left|\frac{dz}{z_0}\right| \le \left|\frac{f_x'(x_0,y_0)}{f(x_0,y_0)}\right|\delta_1 + \left|\frac{f_y'(x_0,y_0)}{f(x_0,y_0)}\right|\delta_2.$$

P.83 定理 8.7 若函数$u=\varphi(x,y), v=\psi(x,y)$在点$(x,y)$处偏导数都存在，$z=f(u,v)$在点$(u,v)=(\varphi(x,y),\psi(x,y))$处可微，则复合函数$z=f[\varphi(x,y),\psi(x,y)]$.在点$(x,y)$处的偏导数存在，并有下列的求导公式

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}, \quad \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y} \quad (8.3)$$

提示：先证明(8.3)第一个公式，故设$\Delta y = 0, \Delta x \ne 0$,得到$u, v$对$x$的偏增量

$\Delta_x u = \varphi(x+\Delta x, y) - \varphi(x,y);\ \Delta_x v = \psi(x+\Delta x, y) - \psi(x,y).$

由$z=f(u,v)$在点(u,v)处可微，由全增量公式

$$\Delta z = \frac{\partial z}{\partial u}\cdot\Delta u + \frac{\partial z}{\partial v}\cdot\Delta v + \varepsilon_1\Delta u + \varepsilon_2\Delta v,$$

其中：$\lim\limits_{\substack{\Delta u\to 0\\ \Delta v\to 0}} \varepsilon_1 = 0, \lim\limits_{\substack{\Delta u\to 0\\ \Delta v\to 0}} \varepsilon_2 = 0$,把$\Delta_x u, \Delta_x v$代入上式，

得到z对x的偏增量：$\Delta_x z = \frac{\partial z}{\partial u}\cdot\Delta_x u + \frac{\partial z}{\partial v}\cdot\Delta_x v + \varepsilon_1\Delta_x u + \varepsilon_2\Delta_x v.$

$\because \Delta x \ne 0$ 得 $\frac{\Delta_x z}{\Delta x} = \frac{\partial z}{\partial u}\cdot\frac{\Delta_x u}{\Delta x} + \frac{\partial z}{\partial v}\cdot\frac{\Delta_x v}{\Delta x} + \varepsilon_1\frac{\Delta_x u}{\Delta x} + \varepsilon_2\frac{\Delta_x v}{\Delta x}$

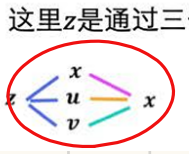
$\lim\limits_{\Delta x\to 0}\frac{\Delta_x u}{\Delta x} = \frac{\partial u}{\partial x} \Rightarrow \Delta_x u \to 0 (\Delta x \to 0)$　　$\lim\limits_{\Delta x\to 0}\frac{\Delta_x v}{\Delta x} = \frac{\partial v}{\partial x} \Rightarrow \Delta_x v \to 0 (\Delta x \to 0)$

$$\begin{cases}\Delta_x u \to 0\\ \Delta_x v \to 0\\ \Delta y = 0\\ \Delta x \to 0\end{cases} \Rightarrow \begin{cases}\Delta u \to 0\\ \Delta v \to 0\end{cases} \Rightarrow \begin{cases}\varepsilon_1 \to 0\\ \varepsilon_2 \to 0\end{cases} \Rightarrow \lim_{\Delta x\to 0}\frac{\Delta_x z}{\Delta x} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x} = \frac{\partial z}{\partial x}$$

同理证明(8.3)第二个公式。

注：（1）由于直接验证$z=f(u,v)$可微不容易，有时假设$f(u,v)$是有连续的偏导数（由微分的充分条件知$f(u,v)$可微）。

（2）设$z=f(x,u,v), u=u(x), v=v(x)$复合结构如右图，这里$z$是通过三个中间变量是$x$的一元复合函数,则



P.85 例 2 $z=f(x^2-y^2, xy)$,其中f有二阶连续偏导数,求 $\frac{\partial^2 z}{\partial x\partial y}$.

提示：为简便起见，不再引入中间变量u, v的记号，约定：

f_1'表示对第一个中间变量求偏导

f_2'表示对第二个中间变量求偏导，

f_{12}''表示先对第一个中间变量求偏导再对第二个中间变量求偏导，上式可写为：

$$\frac{\partial z}{\partial x} = f_1'\cdot 2x + f_2'\cdot y;$$

$$\frac{\partial^2 z}{\partial x\partial y} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial y}(f_1'\cdot 2x + f_2'\cdot y) = 2x\left(\frac{\partial}{\partial y}f_1'\right) + \frac{\partial}{\partial y}(f_2'\cdot y)$$

$$= 2x[f_{11}''(-2y) + f_{12}''x] + [f_{21}''(-2y) + f_{22}''x]y + f_2'$$

$$= -4xyf_{11}'' + 2(x^2-y^2)f_{12}'' + xyf_{22}'' + f_2'.$$

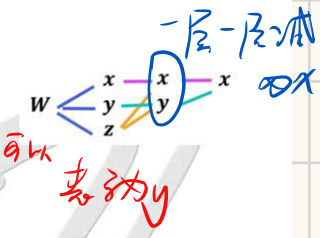
注意：这里 $f_{12}'' = f_{21}''$

$$\frac{dz}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial u}\cdot\frac{du}{dx} + \frac{\partial f}{\partial v}\cdot\frac{dv}{dx} \quad (\text{全导数}).$$

所有变量都与x有关

例如：P.89 8-3

2. 设函数$W=F(x,y,z)$，$z=f(x,y)$，$y=\varphi(x)$，其中F、f具有连续的偏导数，φ可导，求$\frac{dW}{dx}$.



y之都与x有关 表为y

解：$\frac{dW}{dx} = F_x' + F_y'\varphi' + F_z'[f_x' + f_y'\cdot\varphi']$ （全导数）

（3）定理 8.7 要求外函数f在(u,v)处可微是必须的，否则定理结论不一定成立。反例：

$$f(x,y) = \begin{cases}\frac{x^2y}{x^2+y^2}, & (x^2+y^2)\ne 0\\ 0, & (x^2+y^2)=0\end{cases}$$

计算得：$f_x'(0,0) = f_y'(0,0) = 0.$

$$\lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\frac{\Delta z - [f_x'(0,0)\Delta x + f_y'(0,0)\Delta y]}{\rho} = \lim_{\substack{\Delta x\to 0\\ \Delta y\to 0}}\frac{(\Delta x)^2\Delta y}{[(\Delta x)^2+(\Delta y)^2]^{\frac{3}{2}}}$$

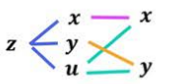
$$\lim_{\substack{\Delta x\to 0\\ \Delta y = k\Delta x}}\frac{(\Delta x)^2\Delta y}{[(\Delta x)^2+(\Delta y)^2]^{\frac{3}{2}}} = \frac{k}{(1+k^2)^{\frac{3}{2}}} \Rightarrow f\text{在}(0,0)\text{处不可微}$$

另一方面：

$$\begin{cases}x=t\\ y=t\end{cases} \Rightarrow u \triangleq f(x,y) = f(t,t) \triangleq g(t) = \frac{t}{2}\ (t\in\mathbb{R}) \Rightarrow \frac{du}{dt} = \frac{1}{2}$$

而链式法则得：$\frac{du}{dt} = \frac{\partial f}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial f}{\partial y}\cdot\frac{dy}{dt} = 0.$

例 设$z=f(x,y,u), u=u(x,y)$,请问：$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x}$?



答:$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x}\cdot\frac{dx}{dx} + \frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial x} + \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x}$

注意：这里$\frac{\partial z}{\partial x}$与$\frac{\partial f}{\partial x}$是两个不同的概念：

$\frac{\partial z}{\partial x}$是复合函数,$z=f[x,y,u(x,y)]$对自变量$x$的偏导，

而$\frac{\partial f}{\partial x}$是外函数,$f(x,y,u)$对中间变量$x$求偏导,这时$y,u$都看成常数.

因此，右边的$\frac{\partial f}{\partial x}$不要写成$\frac{\partial z}{\partial x}$,以免引起混淆.

同理 $\frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} + \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y}$

复合函数求导例题

P.86 例 3 设 $u=f(x+y+z,x^2+y^2+z^2)$,其中f有二阶连续偏导数,

求 $\Delta u=\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}+\dfrac{\partial^2 u}{\partial z^2}$.

f'仍有两变量

提示: 由 $\dfrac{\partial u}{\partial x}=f_1'+f_2'\cdot 2x$, 复合函数求导

$$\frac{\partial^2 u}{\partial x^2}=f_{11}''+f_{12}''2x+2(f_{21}''+f_{22}''2x)x+2f_2'=f_{11}''+4xf_{12}''+4x^2f_{22}''+2f_2'$$

由对称性得

$$\frac{\partial^2 u}{\partial y^2}=f_{11}''+4yf_{12}''+4y^2f_{22}''+2f_2'$$

$$\frac{\partial^2 u}{\partial z^2}=f_{11}''+4zf_{12}''+4z^2f_{22}''+2f_2'$$

于是 $\Delta u=3f_{11}''+4(x+y+z)f_{12}''+4(x^2+y^2+z^2)f_{22}''+6f_2'$

补充题 设$z=f\left(\dfrac{y}{x},xe^y\right)$,且$f$具有二阶连续偏导数,求:$\dfrac{\partial^2 z}{\partial x\partial y}$

解: $\dfrac{\partial z}{\partial x}=-\dfrac{y}{x^2}f_1'+e^yf_2'$; $\dfrac{\partial z}{\partial y}=\dfrac{1}{x}f_1'+xe^yf_2'$

$$\frac{\partial^2 z}{\partial x\partial y}=-\frac{1}{x^2}f_1'-\frac{y}{x^2}\left(\frac{1}{x}f_{11}''+xe^yf_{12}''\right)+e^yf_2'+e^y\left(\frac{1}{x}f_{21}''+xe^yf_{22}''\right)$$

$$=-\frac{1}{x^2}f_1'+e^yf_2'-\frac{y}{x^3}f_{11}''+\frac{e^y(1-y)}{x}f_{12}''+xe^{2y}f_{22}''$$

补充题: 设函数$u=u(x,y)$具有连续的二阶偏导数,满足方程

$$\frac{\partial^2 u}{\partial x^2}+x\frac{\partial^2 u}{\partial x\partial y}+y\frac{\partial^2 u}{\partial y^2}+\frac{\partial u}{\partial y}=xy$$

作自变量变换$x=\xi+\eta$, $y=\xi\eta$,证明:

$$\frac{\partial^2 u}{\partial \xi\partial \eta}=\xi^2\eta+\xi\eta^2$$

证明:

u_1'仍是xy函数

$u_{11}'\cdot 1+u_{12}'\cdot\xi$

$\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 v}{\partial x\partial y}$

$$\begin{cases}x=\xi+\eta\\ y=\xi\eta\end{cases}\Rightarrow\begin{cases}dx=d\xi+d\eta\\ dy=\xi d\eta+\eta d\xi\end{cases}$$

$$\frac{\partial^2 u}{\partial\xi\partial\eta}=\frac{\partial}{\partial\eta}\left(\frac{\partial u}{\partial\xi}\right)=\frac{\partial}{\partial\eta}\left(\frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial\xi}+\frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial\xi}\right)=\frac{\partial}{\partial\eta}\left(\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}\cdot\eta\right)=\frac{\partial}{\partial\eta}\left(\frac{\partial u}{\partial x}\right)+\frac{\partial}{\partial\eta}\left(\frac{\partial u}{\partial y}\cdot\eta\right)$$

$$=\left[\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)\cdot\frac{\partial x}{\partial\eta}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)\cdot\frac{\partial y}{\partial\eta}\right]+\frac{\partial}{\partial\eta}\left(\frac{\partial u}{\partial y}\right)\cdot\eta+\frac{\partial u}{\partial y}$$

$u_{11}'\eta+u_{12}'\xi\eta+u_{21}'$

$$=\left[\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right)\cdot\frac{\partial x}{\partial\eta}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right)\cdot\frac{\partial y}{\partial\eta}\right]+\left[\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right)\frac{\partial x}{\partial\eta}+\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right)\frac{\partial y}{\partial\eta}\right]\cdot\eta+\frac{\partial u}{\partial y}$$

$$=\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial x\partial y}\xi+\frac{\partial^2 u}{\partial y\partial x}\eta+\frac{\partial^2 u}{\partial y^2}\xi\eta+\frac{\partial u}{\partial y}=\frac{\partial^2 u}{\partial x^2}+(\xi+\eta)\frac{\partial^2 u}{\partial x\partial y}+\xi\eta\frac{\partial^2 u}{\partial y^2}+\frac{\partial u}{\partial y}$$

$$=\frac{\partial^2 u}{\partial x^2}+(\xi+\eta)\frac{\partial^2 u}{\partial x\partial y}+\xi\eta\frac{\partial^2 u}{\partial y^2}+\frac{\partial u}{\partial y}=\frac{\partial^2 u}{\partial x^2}+x\frac{\partial^2 u}{\partial x\partial y}+y\frac{\partial^2 u}{\partial y^2}+\frac{\partial u}{\partial y}=xy$$

$$=(\xi+\eta)\xi\eta=\xi^2\eta+\xi\eta^2$$

复合函数求导麻烦容易错,因此引入

一阶微分不变性

$z=z(x,y)\Rightarrow dz=\dfrac{\partial z}{\partial x}dx+\dfrac{\partial z}{\partial y}dy$

复合函数$z=z(x(s,t),y(s,t))$可微，且$dz=\frac{\partial z}{\partial s}ds+\frac{\partial z}{\partial t}dt$,

用复合函数求导法则

$$\begin{cases}\frac{\partial z}{\partial s}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial s}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial s}\\ \frac{\partial z}{\partial t}=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial t}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial t}\end{cases}\Rightarrow dz=\left(\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial s}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial s}\right)ds+\left(\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial t}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial t}\right)dt$$

即dx

$$=\frac{\partial z}{\partial x}\left(\frac{\partial x}{\partial s}ds+\frac{\partial x}{\partial t}dt\right)+\frac{\partial z}{\partial y}\left(\frac{\partial y}{\partial s}ds+\frac{\partial y}{\partial t}dt\right)=\frac{\partial z}{\partial x}dx+\frac{\partial z}{\partial y}dy.$$

* 若是 s,t 函数，依旧可以化简到 dx

换句话说，若$z=z(u,v)$，且$dz=\varphi(u,v)du+\psi(u,v)dv$，则

$$\frac{\partial z}{\partial u}=\varphi(u,v),\frac{\partial z}{\partial v}=\psi(u,v).$$

注意：(1) 一元、多元函数都不具有高阶微分不变性.

(2)当x,y是自变量时，dx和dy各独立取值，

当x,y是中间变量时，它们的取值由s,t,ds,dt确定.

微分一阶形式不变性和微分的四则运算提供另一个的计算复杂函数的全微分及偏导数的方法：

P.88 例 7 设$z=f(r\cos\theta,r\sin\theta)$，求$dz$，并由此求$\frac{\partial z}{\partial r},\frac{\partial z}{\partial\theta}$.

解: $dz=df(r\cos\theta,r\sin\theta)=f_1'd(r\cos\theta)+f_2'd(r\sin\theta)$

$=f_1'(\cos\theta dr+rd\cos\theta)+f_2'(\sin\theta dr+rd\sin\theta)$

$=f_1'(\cos\theta dr-r\sin\theta d\theta)+f_2'(\sin\theta dr+r\cos\theta d\theta)$

$=(f_1'\cos\theta+f_2'\sin\theta)dr+(-f_1'r\sin\theta+f_2'r\cos\theta)d\theta$

由此得到$\frac{\partial z}{\partial r}=f_1'\cos\theta+f_2'\sin\theta,\quad\frac{\partial z}{\partial\theta}=-f_1'r\sin\theta+f_2'r\cos\theta$.

表示对第一个变量求导

补充题：设$z=f(xy,e^{x-2y})$，且f具有一阶连续偏导，求:$\frac{\partial z}{\partial x},\frac{\partial z}{\partial y}$

解: $dz=f_1'd(xy)+f_2'd(e^{x-2y})=f_1'(ydx+xdy)+e^{x-2y}f_2'(dx-2dy)$

$=(f_1'y+e^{x-2y}f_2')dx+(xf_1'-2e^{x-2y}f_2')dy$

$\Rightarrow\frac{\partial z}{\partial x}=f_1'y+e^{x-2y}f_2',\quad\frac{\partial z}{\partial y}=xf_1'-2e^{x-2y}f_2'$

第四节 隐函数的偏导数

定理 8.8' 若函数$F(x,y)$满足下列条件：

(1) 函数F在以$P_0(x_0,y_0)$为内点的某一区域$D\subseteq R^2$上连续；

(2) $F(x_0,y_0)=0$ (初始条件)

(3) 在D内存在连续的偏导数$F_x'(x,y),F_y'(x,y)$;

(4) $F_y'(x_0,y_0)\neq0$

则在点P_0的某邻域$U(P_0)$内唯一确定一个具有连续导数的函数

二元： $y=f(x)$ $F(x,y(x))=0$

$$\Rightarrow \frac{\partial y}{\partial x}=-\frac{F_x'}{F_y'}$$

三元

P.90 定理 8.8 若函数$F(x,y,z)$满足下列条件：

(1) 函数F在以$P_0(x_0,y_0,z_0)$为内点的某一区域$D\subseteq R^3$上连续；

(2) $F(x_0,y_0,z_0)=0$（初始条件）

(3) 在D内存在连续的偏导数$F_x'(x,y,z),F_y'(x,y,z),F_z'(x,y,z)$；

(4) $F_z'(x_0,y_0,z_0)\neq 0$

$\because F(x,y,z(x,y))=0$ 左右对x求偏导

$$F_x'\cdot\frac{\partial x}{\partial x}+F_z'\frac{\partial z}{\partial x}=0$$

$$\therefore \frac{\partial z}{\partial x}=-\frac{F_x'}{F_z}$$

同理 $\frac{\partial z}{\partial y}=-\frac{F_y'}{F_z'}$

条件 一阶微分不变性

$F(x,y,z(x,y))=0$ $F_z'\neq 0$

$$dF=F_x'dx+F_y'dy+F_z'dz=0$$

$$\Rightarrow dz=-\frac{F_x'}{F_z'}dx-\frac{F_y'}{F_z'}dy$$

$$\frac{\partial z}{\partial x}=-\frac{F_x'}{F_z'}\qquad \frac{\partial z}{\partial y}=-\frac{F_y'}{F_z'}$$

(2) 注意:方程$F(x,y,z)=0,F_x'\neq 0,F_y'\neq 0,F_z'\neq 0\left(\Leftrightarrow F_x'F_y'F_z'\neq 0\right)$

有：$dz=-\frac{F_x'}{F_z'}dx-\frac{F_y'}{F_z'}dy\Rightarrow\frac{\partial z}{\partial x}=-\frac{F_x'}{F_z'}$

$dy=-\frac{F_x'}{F_y'}dx-\frac{F_z'}{F_y'}dz\Rightarrow\frac{\partial y}{\partial z}=-\frac{F_z'}{F_y'}\Rightarrow\frac{\partial x}{\partial y}\cdot\frac{\partial y}{\partial z}\cdot\frac{\partial z}{\partial x}=-1$

$dx=-\frac{F_y'}{F_x'}dy-\frac{F_z'}{F_x'}dz\Rightarrow\frac{\partial x}{\partial y}=-\frac{F_y'}{F_x'}$

则得$\frac{\partial z}{\partial x},\frac{\partial y}{\partial z},\frac{\partial z}{\partial x}$中的$\partial x,\partial y,\partial z$不可分割。

(3) $\frac{\partial z}{\partial x}=-\frac{F_x'}{F_z'},\frac{\partial z}{\partial y}=-\frac{F_y'}{F_z'}$，仅适用于$F(x,y,z)=0$，对于显式$z=z(x,y)$不使用，

显式只能用复合函数求偏导或一阶微分不变性来求。

F是含 z 的函数，z并没拆成 x,y

问：满足什么条件下$F(2x+y^2-z,xyz)=0$确定唯一的一个隐函数$z=z(x,y)$?

答：$G(x,y,z)\triangleq F(2x+y^2-z,xyz)=0$ 满足定理 8.8 的定理条件

$\Leftrightarrow$ 存在某邻域内$G_z'(x,y,z)\neq 0$。

$G(x,y,z)=F(2x+y^2-z,xyz)\Rightarrow G_z'=-F_1'+F_2'\cdot(xy)\Rightarrow F_1'-xyF_2'\neq 0$

法一：复合函数求偏导：

$$F(2x+y^2-z,xyz)=0\Rightarrow F_1'\cdot\left(2-\frac{\partial z}{\partial x}\right)+F_2'\cdot\left(yz+xy\frac{\partial z}{\partial x}\right)=0$$

$$\Rightarrow\frac{\partial z}{\partial x}=\frac{2F_1'+xzF_2'}{F_1'-xyF_2'}$$

同理得：$\dfrac{\partial z}{\partial x}=\dfrac{2yF_1'+yzF_2'}{F_1'-xyF_2'}$

法二：一阶微分不变性：(推荐)

$$F(2x+y^2-z,xyz)=0\Rightarrow 0=dF=F_1'\cdot d(2x+y^2-z)+F_2'\cdot d(xyz)$$

$$=(2F_1'+yzF_2')dx+(2yF_1'+xzF_2')dy-(F_1'-xyF_2')dz=0$$

$$\Rightarrow\begin{cases}\dfrac{\partial z}{\partial x}=\dfrac{2F_1'+xzF_2'}{F_1'-xyF_2'}\\[2ex]\dfrac{\partial z}{\partial y}=\dfrac{2yF_1'+yzF_2'}{F_1'-xyF_2'}\end{cases}(F_1'-xyF_2'\neq 0)$$

补充题 1 设$z=z(x,y)$是由$e^{x+z}-2xz+y=e$确定的隐函数，求：$\dfrac{\partial z}{\partial x},\dfrac{\partial^2 z}{\partial x\partial y}\Big|_{\substack{x=0\\y=0}}$

解：

方法一 用一阶微分不变性来求解。

方程$e^{x+z}-2xz+y=e$两边分别微分得：

$e^{x+z}(dx+dz)-2zdx-2xdz+dy=0$

$\Rightarrow(e^{x+z}-2z)dx+dy+(e^{x+z}-2x)dz=0$

$$\Rightarrow(e^{x+z}-2x)dz=(2z-e^{x+z})dx-dy\Rightarrow\begin{cases}\dfrac{\partial z}{\partial x}=\dfrac{2z-e^{x+z}}{e^{x+z}-2x}\\[2ex]\dfrac{\partial z}{\partial y}=-\dfrac{1}{e^{x+z}-2x}\end{cases}$$

$$\text{当}\begin{cases}x=0\\y=0\end{cases}\Rightarrow z=1\Rightarrow\begin{cases}\left.\dfrac{\partial z}{\partial x}\right|_{\substack{x=0\\y=0}}=2e^{-1}-1\\[2ex]\left.\dfrac{\partial z}{\partial y}\right|_{\substack{x=0\\y=0}}=-e^{-1}\end{cases}$$

$$\frac{\partial^2 z}{\partial x\partial y}=\frac{\partial}{\partial y}\left(\frac{2z-e^{x+z}}{e^{x+z}-2x}\right)=\frac{\left(2\dfrac{\partial z}{\partial y}-e^{x+z}\dfrac{\partial z}{\partial y}\right)(e^{x+z}-2x)-(2z-e^{x+z})\left(e^{x+z}\dfrac{\partial z}{\partial y}\right)}{(e^{x+z}-2x)^2}$$

$$\left.\frac{\partial^2 z}{\partial x\partial y}\right|_{\substack{x=0\\y=0}}=\left.\frac{\left(2\dfrac{\partial z}{\partial y}-e^{x+z}\dfrac{\partial z}{\partial y}\right)(e^{x+z}-2x)-(2z-e^{x+z})\left(e^{x+z}\dfrac{\partial z}{\partial y}\right)}{(e^{x+z}-2x)^2}\right|_{\substack{x=0\\y=0}}$$

$$\text{当}\begin{cases}x=0\\y=0\end{cases}\Rightarrow z=1\Rightarrow\begin{cases}\left.\dfrac{\partial z}{\partial x}\right|_{\substack{x=0\\y=0}}=2e^{-1}-1\\[2ex]\left.\dfrac{\partial z}{\partial y}\right|_{\substack{x=0\\y=0}}=-e^{-1}\end{cases}$$

对$e^{x+z}\cdot\left(1+\dfrac{\partial z}{\partial x}\right)-2z-2x\dfrac{\partial z}{\partial x}=0$ 两边再对y求偏导，有：

$$e^{x+z}\cdot\frac{\partial z}{\partial y}\cdot\left(1+\frac{\partial z}{\partial x}\right)-e^{x+z}\cdot\frac{\partial^2 z}{\partial x\partial y}-2\frac{\partial z}{\partial y}-2x\frac{\partial^2 z}{\partial x\partial y}=0\Rightarrow\left.\frac{\partial^2 z}{\partial x\partial y}\right|_{\substack{x=0\\y=0}}=0$$

隐函数组的偏导数

P.92 定理 8.9 隐函数组存在定理

设函数$F(x,y,u,v),G(x,y,u,v)$在点$P_0(x_0,y_0,u_0,v_0)$的某邻域内满足:

(1) $F(x_0,y_0,u_0,v_0)=G(x_0,y_0,u_0,v_0)=0$(初始条件)

(2) F,G存在一阶连续偏导数$F'_x,F'_y,F'_u,F'_v,G'_x,G'_y,G'_u,G'_v$,

(3)在点P_0处行列式$J\triangleq\dfrac{\partial(F,G)}{\partial(u,v)}=\begin{vmatrix}F'_u & F'_v\\ G'_u & G'_v\end{vmatrix}\neq 0$(*Jacobi*行列式)

则方程组$\begin{cases}F(x,y,u,v)=0\\ G(x,y,u,v)=0\end{cases}$在点$P_0(x_0,y_0,u_0,v_0)$的某邻域内唯一确定一组函数组的$\begin{cases}u=u(x,y)\\ v=v(x,y)\end{cases}$满足:

(1) $u_0=u(x_0,y_0),\ v_0=v(x_0,y_0)$;

(2) $F(x,y,u(x,y),v(x,y))=0,G(x,y,u(x,y),v(x,y))=0$;

(3)$u=u(x,y),v=v(x,y)$具有一阶连续偏导数:

$$\frac{\partial u}{\partial x}=-\frac{1}{J}\cdot\frac{\partial(F,G)}{\partial(x,v)}=-\frac{\begin{vmatrix}F'_x & F'_v\\ G'_x & G'_v\end{vmatrix}}{\begin{vmatrix}F'_u & F'_v\\ G'_u & G'_v\end{vmatrix}},\frac{\partial v}{\partial x}=-\frac{1}{J}\cdot\frac{\partial(F,G)}{\partial(u,x)}=-\frac{\begin{vmatrix}F'_u & F'_x\\ G'_u & G'_x\end{vmatrix}}{\begin{vmatrix}F'_u & F'_v\\ G'_u & G'_v\end{vmatrix}},$$

$$\frac{\partial u}{\partial y}=-\frac{1}{J}\cdot\frac{\partial(F,G)}{\partial(y,v)}=-\frac{\begin{vmatrix}F'_y & F'_v\\ G'_y & G'_v\end{vmatrix}}{\begin{vmatrix}F'_u & F'_v\\ G'_u & G'_v\end{vmatrix}},\frac{\partial v}{\partial y}=-\frac{1}{J}\cdot\frac{\partial(F,G)}{\partial(u,y)}=-\frac{\begin{vmatrix}F'_u & F'_y\\ G'_u & G'_y\end{vmatrix}}{\begin{vmatrix}F'_u & F'_v\\ G'_u & G'_v\end{vmatrix}}.$$

其中$J=\begin{vmatrix}F'_u & F'_v\\ G'_u & G'_v\end{vmatrix}\triangleq\dfrac{\partial(F,G)}{\partial(u,v)}\neq 0$ 称为函数F,G的*Jacobi*行列式.

雅可比行列式

$$\frac{\partial(F,G)}{\partial(u,v)}=\begin{vmatrix}F_u' & G_u'\\ F_v' & G_v'\end{vmatrix}$$

$F(x,y,u,v)=0$

$G(x,y,u,v)=0$

$u=(x,y)\quad v=v(x,y)$

$F(x,y,u(x,y),v(x,y))\equiv 0$

$G(x,y,u(x,y),v(x,y))\equiv 0$

$F_x'+F_u'\cdot\dfrac{\partial u}{\partial x}+F_v'\dfrac{\partial v}{\partial x}=0$

$G_x'+G_u'\cdot\dfrac{\partial u}{\partial x}+G_v'\dfrac{\partial v}{\partial x}=0$

$$\begin{pmatrix}F_u' & F_v'\\ G_u' & G_v'\end{pmatrix}\begin{pmatrix}\frac{\partial u}{\partial x}\\ \frac{\partial v}{\partial x}\end{pmatrix}=\begin{pmatrix}-F_x'\\ -G_x'\end{pmatrix}$$

P.94 例 4 设$\begin{cases}u^2+v^2-x^2-y=0\\ -u+v-xy+1=0\end{cases}$,求$\dfrac{\partial x}{\partial u},\dfrac{\partial y}{\partial u}$. 复合函数求导

解 $\begin{cases}u^2+v^2-x^2-y=0\\ -u+v-xy+1=0\end{cases}\Rightarrow\begin{cases}2u-2x\cdot\dfrac{\partial x}{\partial u}-\dfrac{\partial y}{\partial u}=0\\ -1-\dfrac{\partial x}{\partial u}\cdot y-x\dfrac{\partial y}{\partial u}=0\end{cases}\Rightarrow\begin{cases}\dfrac{\partial x}{\partial u}=\dfrac{2xu+1}{2x^2-y}\\ \dfrac{\partial y}{\partial u}=-\dfrac{2x+2yu}{2x^2-y}\end{cases}$

P.94 例 5 设$\begin{cases}xu-yv=0\\ yu+xv=1\end{cases}$,求$\dfrac{\partial u}{\partial x},\dfrac{\partial u}{\partial y},\dfrac{\partial v}{\partial x},\dfrac{\partial v}{\partial y}$.

一阶微分不变性

解 $\begin{cases}xu-yv=0\\ yu+xv=1\end{cases}\overset{\text{微分}}{\Rightarrow}\begin{cases}xdu+udx-ydv-vdy=0\\ ydu+udy+xdv+vdx=0\end{cases}$

$$\Rightarrow\begin{cases}du=\dfrac{1}{x^2+y^2}[-(xu+yv)dx+(xv-yu)dy]\\ dv=\dfrac{1}{x^2+y^2}[(yu-xv)dx-(xu+yv)dy]\end{cases}$$

$$\Rightarrow\begin{cases}\dfrac{\partial u}{\partial x}=-\dfrac{xu+yv}{x^2+y^2}\\ \dfrac{\partial u}{\partial y}=\dfrac{xv-yu}{x^2+y^2}\end{cases},\begin{cases}\dfrac{\partial v}{\partial x}=\dfrac{yu-xv}{x^2+y^2}\\ \dfrac{\partial v}{\partial y}=-\dfrac{xu+yv}{x^2+y^2}\end{cases}.$$

全导数(抓住一个自变量)

P.94 例 6 设$u=f(x,y,z),\psi(x^2,e^y,z)=0,y=\sin x$,其中$f,\psi$具有连续的偏导数且$\psi_3'\neq 0$,求$\dfrac{du}{dx}$.

一阶微分不变性

x — x — x
y — y
z

解 法一

$$u=f(x,y,z)\Rightarrow\frac{du}{dx}=f_x'+f_y'\frac{dy}{dx}+f_z'\frac{dz}{dx}\qquad(1)$$

$$y=\sin x\Rightarrow\frac{dy}{dx}=\cos x\qquad(2)$$

$$\psi(x^2,e^y,z)=0\Rightarrow\psi_1'\cdot 2x+\psi_2'\cdot e^y\cdot\frac{dy}{dx}+\psi_3'\cdot\frac{dz}{dx}=0$$

$$\Rightarrow\frac{dz}{dx}=-\frac{1}{\psi_3'}(\psi_1'\cdot 2x+e^y\cdot\cos x\cdot\psi_2')\qquad(3)$$

把(2),(3)代入(1):

$$\frac{du}{dx}=f_x'+f_y'\cos x-f_z'\frac{1}{\psi_3'}(2x\psi_1'+e^y\cos x\psi_2').$$

法二 一阶微分不变性+微分四则运算

$$u=f(x,y,z)\Rightarrow du=f_x'dx+f_y'dy+f_z'dz\qquad(4)$$

$$\psi(x^2,e^y,z)=0\Rightarrow\psi_1'd(x^2)+\psi_2'd(e^y)+\psi_3'dz=0\qquad(5)$$

$$\Rightarrow dz=-\frac{1}{\psi_3'}(\psi_1'2x+\psi_2'e^y\cos x)dx\qquad(6)$$

(5),(6) 代入(4): $du=\left[f_x'+f_y'\cos x-\dfrac{f_z'}{\psi_3'}(\psi_1'2x+\psi_2'e^y\cos x)\right]dx$

$$\Leftrightarrow\frac{du}{dx}=f_x'+f_y'\cos x-\frac{f_z'}{\psi_3'}(\psi_1'2x+\psi_2'e^y\cos x)$$

反函数的偏导数

$$\begin{cases} x = x(u,v) \\ y = y(u,v) \end{cases} \Longleftrightarrow \text{反函数组} \quad \begin{cases} u = u(x,y) \\ v = v(x,y) \end{cases}$$

$$\begin{cases} x - x[u(x,y), v(x,y)] = 0 \\ y - y[u(x,y), v(x,y)] = 0 \end{cases}$$

对x求导

$$\begin{cases} 1 - x_u' u_x' - x_v' v_x' = 0 \\ 0 - y_u' u_x' - y_v' v_x' = 0 \end{cases} \qquad \begin{cases} x_u' u_x' + x_v' v_x' = 1 \\ y_u' u_x' + y_v' v_x' = 0 \end{cases}$$

对y求导

$$\begin{cases} 0 - x_u' u_y' - x_v' v_y' = 0 \\ 1 - y_u' u_y' - x_v' v_y' = 0 \end{cases} \qquad \begin{cases} x_u' u_y' + x_v' v_y' = 0 \\ y_u' u_y' + x_v' v_y' = 1 \end{cases}$$

$$\frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} u_x' & v_x' \\ u_y' & v_y' \end{vmatrix} \cdot \begin{vmatrix} x_u' & y_u' \\ x_v' & y_v' \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$$

与反函数 $\frac{dy}{dx} \cdot \frac{dx}{dy} = 1$ 一致

推广

$$\begin{cases} x = x(u,v,w) \\ y = y(u,v,w) \\ z = z(u,v,w) \end{cases} \Longleftrightarrow \begin{cases} u = u(x,y,z) \\ v = v(x,y,z) \\ w = w(x,y,z) \end{cases} \Longrightarrow \frac{\partial(u,v,w)}{\partial(x,y,z)} \cdot \frac{\partial(x,y,z)}{\partial(u,v,w)} = 1$$

例 7 $\begin{cases} x = e^u + u\sin v \\ y = e^u - u\cos v \end{cases}$ 确定反函数组 $\begin{cases} u = u(x,y) \\ v = v(x,y) \end{cases}$，求 $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$.

解：$\begin{cases} x = e^u + u\sin v \\ y = e^u - u\cos v \end{cases} \Rightarrow \begin{cases} \begin{cases} 1 = e^u \cdot \frac{\partial u}{\partial x} + \frac{\partial u}{\partial x}\sin v + u\cos v \frac{\partial v}{\partial x} \\ 0 = e^u \cdot \frac{\partial u}{\partial x} - \frac{\partial u}{\partial x}\cos v + u\sin v \frac{\partial v}{\partial x} \end{cases} \\ \begin{cases} 0 = e^u \cdot \frac{\partial u}{\partial y} + \frac{\partial u}{\partial y}\sin v + u\cos v \frac{\partial v}{\partial y} \\ 1 = e^u \cdot \frac{\partial u}{\partial y} - \frac{\partial u}{\partial y}\cos v + u\sin v \frac{\partial v}{\partial y} \end{cases} \end{cases}$

$$\Rightarrow \begin{cases} \frac{\partial u}{\partial x} = \frac{\sin v}{e^u(\sin v - \cos v) + 1} \\ \frac{\partial v}{\partial x} = \frac{-e^u + \cos v}{u[e^u(\sin v - \cos v) + 1]} \\ \frac{\partial u}{\partial y} = \frac{-\cos v}{e^u(\sin v - \cos v) + 1} \\ \frac{\partial v}{\partial y} = \frac{e^u + \sin v}{u[e^u(\sin v - \cos v) + 1]} \end{cases}$$

复合函数求导

例 8 设 $\begin{cases} x = -u^2 + v \\ y = u + v^2 \end{cases}$ 确定反函数组 $\begin{cases} u = u(x,y) \\ v = v(x,y) \end{cases}$ 求 $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$.

解：$\begin{cases} x = -u^2 + v \\ y = u + v^2 \end{cases} \Rightarrow \begin{cases} \begin{cases} 1 = -2u \cdot \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x} \\ 0 = \frac{\partial u}{\partial x} + 2v\frac{\partial v}{\partial x} \end{cases} \Rightarrow \begin{cases} \frac{\partial u}{\partial x} = \frac{-2v}{4uv+1} \\ \frac{\partial v}{\partial x} = \frac{1}{4uv+1} \end{cases} \\ \begin{cases} 0 = -2u \cdot \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \\ 1 = \frac{\partial u}{\partial y} + 2v\frac{\partial v}{\partial y} \end{cases} \Rightarrow \begin{cases} \frac{\partial u}{\partial y} = \frac{1}{4uv+1} \\ \frac{\partial v}{\partial y} = \frac{2u}{4uv+1} \end{cases} \end{cases}$

场 → 方向导数 → 梯度

数学上一个向量到另一个向量的映射.

方向导数

定义 8.9 对三元函数 u 在 $P_0(x_0,y_0,z_0)$ 某邻域有定义，l 是从 P_0 出发的射线

$P(x,y,z)$ 为 l 上含于 $U(P_0)$ 内任意一点

定义式：$\lim\limits_{\rho\to 0}\dfrac{u(P)-u_0(P)}{\rho}=\lim\limits_{\rho\to 0}\dfrac{\Delta_l u}{\rho}$

ρ 表示 P 与 P_0 两点的距离

称此极限为函数 u 在 P_0 沿方向 $\vec{l}$ 的方向的方向导数，记作 $\left.\dfrac{\partial u}{\partial l}\right|_{P_0}$.

问：偏导数 $\dfrac{\partial u}{\partial x},\dfrac{\partial u}{\partial y}$ 是方向导数吗？

左右导数不一样

答：若 u 在点 P_0 处存在对 x 的偏导数，则

u 在点 P_0 沿 x 轴正方向的方向导数为：$\rho=|\Delta x|=\Delta x$

$$\left.\frac{\partial u}{\partial \vec{l}}\right|_{P_0}=\lim_{\Delta x\to 0^+}\frac{u(x_0+\Delta x,y_0,z_0)-u(x_0,y_0,z_0)}{\Delta x}=\left.\frac{\partial u}{\partial x}\right|_{P_0}$$

u 在点 P_0 沿 x 轴负方向的方向导数为：$\rho=|\Delta x|=-\Delta x$

$$\left.\frac{\partial u}{\partial \vec{l}}\right|_{P_0}=\lim_{\Delta x\to 0^-}\frac{u(x_0+\Delta x,y_0,z_0)-u(x_0,y_0,z_0)}{-\Delta x}=-\left.\frac{\partial u}{\partial x}\right|_{P_0}$$

定理 8.10 若函数 u 在 $P_0(x_0,y_0,z_0)$ 可微，则 u 在点 P_0 处沿任一方向 $\vec{l}$ 的方向导数都存在

且 $\left.\dfrac{\partial u}{\partial l}\right|_{P_0}=\left.\dfrac{\partial u}{\partial x}\right|_{P_0}\cos\alpha+\left.\dfrac{\partial u}{\partial y}\right|_{P_0}\cos\beta+\left.\dfrac{\partial u}{\partial z}\right|_{P_0}\cos\gamma=$

$\left(\left.\dfrac{\partial u}{\partial x}\right|_{P_0},\left.\dfrac{\partial u}{\partial y}\right|_{P_0},\left.\dfrac{\partial u}{\partial z}\right|_{P_0}\right)(\cos\alpha,\cos\beta,\cos\gamma)$

其中 $\vec{l}=(\cos\alpha,\cos\beta\cos\gamma)$

证 $P(x,y,z)$ 为射线 l 上任一点，由 $\overrightarrow{P_0P}=(x-x_0,y-y_0,z-z_0)=(\Delta x,\Delta y,\Delta z)$,

$\overrightarrow{P_0P^0}=\left(\dfrac{\Delta x}{\rho},\dfrac{\Delta y}{\rho},\dfrac{\Delta z}{\rho}\right)=\vec{l}^0$，有 $\dfrac{\Delta x}{\rho}=\cos\alpha,\dfrac{\Delta y}{\rho}=\cos\beta,\dfrac{\Delta z}{\rho}=\cos\gamma$

方向导数

由 $u(P)$ 在点 P_0 可微，则有

可微得到

$$u(P)-u(P_0)=\left.\frac{\partial u}{\partial x}\right|_{P_0}\Delta x+\left.\frac{\partial u}{\partial y}\right|_{P_0}\Delta y+\left.\frac{\partial u}{\partial z}\right|_{P_0}\Delta z+o(\rho)(\rho\to 0)$$

可以写成这样的

$$\lim_{\rho\to 0}\frac{u(P)-u(P_0)}{\rho}=\left[\lim_{\rho\to 0}\left.\frac{\partial u}{\partial x}\right|_{P_0}\frac{\Delta x}{\rho}+\left.\frac{\partial u}{\partial y}\right|_{P_0}\frac{\Delta y}{\rho}+\left.\frac{\partial u}{\partial z}\right|_{P_0}\frac{\Delta z}{\rho}+\frac{o(\rho)}{\rho}\right]$$

可以互相转化

$$=\left.\frac{\partial u}{\partial x}\right|_{P_0}\cos\alpha+\left.\frac{\partial u}{\partial y}\right|_{P_0}\cos\beta+\left.\frac{\partial u}{\partial z}\right|_{P_0}\cos\gamma$$

则：$\left.\dfrac{\partial u}{\partial \vec{l}}\right|_{P_0}=\left.\dfrac{\partial u}{\partial x}\right|_{P_0}\cos\alpha+\left.\dfrac{\partial u}{\partial y}\right|_{P_0}\cos\beta+\left.\dfrac{\partial u}{\partial z}\right|_{P_0}\cos\gamma.$

注：(1)对于二元函数 $z=f(x,y)$:

$$\left.\frac{\partial f}{\partial \vec{l}}\right|_{P_0}=f_x'(x_0,y_0)\cos\alpha+f_y'(x_0,y_0)\cos\beta.$$

其中 α,β 是平面向量 $\vec{l}$ 分别与 x 轴、y 轴正向的夹角.

例如：P.129 综合题 3 函数 $f(x,y)=\sqrt[3]{x^3+y^3}$ 在 $(0,0)$ 处：

分析：设 $\vec{l}^0=(\cos\alpha,\cos\beta)$,

定义对方向导数

$$\lim_{\rho\to 0}\frac{f(x,y)-f(0,0)}{\rho}=\lim_{\rho\to 0}\frac{\sqrt[3]{\rho^3\cos^3\alpha+\rho^3\cos^3\beta}}{\rho}=\sqrt[3]{\cos^3\alpha+\cos^3\beta}=\frac{\partial f}{\partial \vec{l}}$$

$$\lim_{x\to 0}\frac{f(x,0)-f(0,0)}{x}=\lim_{x\to 0}\frac{\sqrt[3]{x^3+0^3}-0}{x}=\lim_{x\to 0}\frac{x}{x}=1=\left.\frac{\partial f}{\partial x}\right|_{(0,0)},$$

偏导数

同理：$\left.\dfrac{\partial f}{\partial y}\right|_{(0,0)}=1$

$\Rightarrow$ 函数 f 在点 $(0,0)$ 处沿任意方向的方向导数及偏导数存在，但

$$\frac{\partial f}{\partial \vec{l}}=\sqrt[3]{\cos^3\alpha+\cos^3\beta}\neq\cos\alpha+\cos\beta=\left(\frac{\partial f}{\partial x},\frac{\partial f}{\partial y}\right)\cdot(\cos\alpha,\cos\beta)$$

其原因：

但不可微

$$\lim_{\rho\to 0}\frac{f(0+\Delta x,0+\Delta y)-f(0,0)-f_x'(0,0)\Delta x-f_y'(0,0)\Delta y}{\sqrt{\Delta x^2+\Delta y^2}}$$

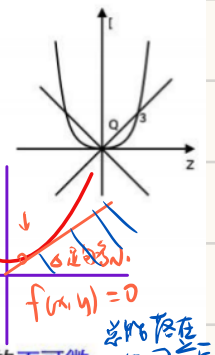
$$=\lim_{\rho\to 0}\frac{\sqrt[3]{\Delta x^3+\Delta y^3}-\Delta x-\Delta y}{\sqrt{\Delta x^2+\Delta y^2}}\overset{\Delta y=k\Delta x,\ \Delta x>0}{=}\lim_{\rho\to 0}\frac{\sqrt[3]{1+k^3}-(k+1)}{\sqrt{1+k^2}}$$

$$=\frac{\sqrt[3]{1+k^3}-(k+1)}{\sqrt{1+k^2}}$$

极限不存在 $\Rightarrow f(x,y)$ 在点 $(0,0)$ 处不可微。

P.101 例 1 求函数$u=\ln(x+\sqrt{y^2+z^2})$在点$A(1,0,1)$处沿点A指向点$B(3,-2,2)$方向的方向导数.(详见 P.101)

P.182 例 2 $f(x,y)=\begin{cases}1, & \text{当 } 0<y<x^2,-\infty<x<+\infty\\ 0, & \text{其余部分.}\end{cases}$



讨论函数f在$(0,0)$处的连续性、偏导数、方向导数及可微性。

解: $\begin{cases}\lim\limits_{\substack{(x,y)\to(0,0)\\0<y<x^2}} f(x,y)=1\\ \lim\limits_{\substack{(x,y)\to(0,0)\\y>x^2}} f(x,y)=0\end{cases}\Rightarrow$ 函数f在$(0,0)$处的不连续.

$\Rightarrow$ 函数f在$(0,0)$处的不可微.

原点出发任何射线都包含原点充分小一段$\Rightarrow$在这一段上$f(x,y)=0$

定义 8.9 $\Rightarrow$ 在原点处沿任何方向$\vec{l}$,都有方向导数$\left.\dfrac{\partial f}{\partial \vec{l}}\right|_{(0,0)}=0$

定义 8.7 $\Rightarrow$ $\begin{cases}\left.\dfrac{\partial f}{\partial x}\right|_{(0,0)}=\lim\limits_{\Delta x\to 0}\dfrac{f(\Delta x,0)-f(0,0)}{\Delta x}=0\\ \left.\dfrac{\partial f}{\partial y}\right|_{(0,0)}=\lim\limits_{\Delta y\to 0}\dfrac{f(0,\Delta y)-f(0,0)}{\Delta y}=0\end{cases}$ 偏导数存在.

结论:

(1) 函数在一点可微是方向导数存在的充分非必要条件;

(2) 函数在一点连续是方向导数存在的非充分非必要条件.

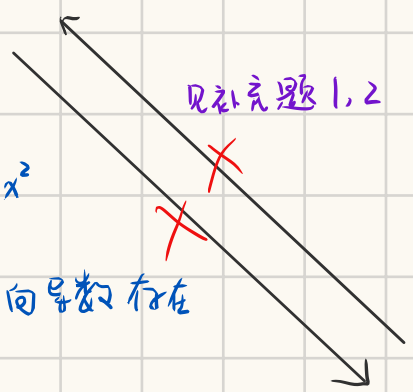
任何方向的方向导数均存在

P100
定理8.10

P102 例2

$f(x)=\begin{cases}1 & 0<y<x^2\\ 0 & \text{其他}\end{cases}$

在(0,0)处不可微,方向导数存在



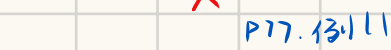
P78
定理 8.6

一阶偏导连续 $\Rightarrow$ 可微 $\Rightarrow$ 一阶偏导存在

P83. 17.

P77
定理 8.5

P77. 例11

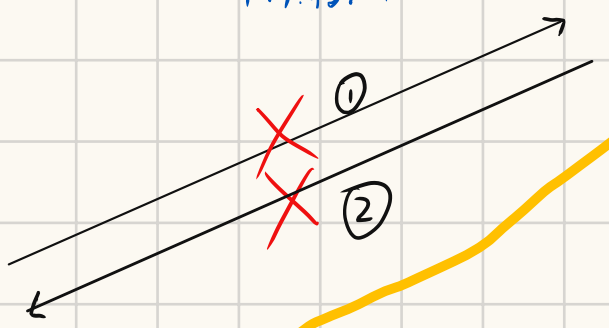


$f(x,y)=\begin{cases}(x^2+y^2)\sin\dfrac{1}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$

可微,一阶偏导不连续

P76
定理 8.4

可微 $\Rightarrow$ 连续



计算方向导数的步骤:

前提:一定要验证定理 8.10 函数u在$P(x_0,y_0,z_0)$可微

step1: 单位化方向 $\vec{l}=(\cos\alpha,\cos\beta,\cos\gamma)$

step2: 计算偏导数 $\dfrac{\partial u}{\partial x},\dfrac{\partial u}{\partial y},\dfrac{\partial u}{\partial z}$ 一阶偏导连续 $\Rightarrow$ 可微 $\Rightarrow$ 定理8.10

step3: 计算方向导数 $\dfrac{\partial u}{\partial l}=\left(\dfrac{\partial u}{\partial x},\dfrac{\partial u}{\partial y},\dfrac{\partial u}{\partial z}\right)(\cos\alpha,\cos\beta,\cos\gamma)$

$=\dfrac{\partial u}{\partial x}\cos\alpha+\dfrac{\partial u}{\partial y}\cos\beta+\dfrac{\partial u}{\partial z}\cos\gamma$

思考：补充题 1 $f(x,y)=\sqrt{x^2+y^2}$

在(0,0)处的任何方向$\vec{l}$的方向导数存在，但是两个偏导数不存在。

证明：函数f在(0,0)处的任何方向$\vec{l}=(\Delta x,\Delta y)$的方向导数：

$$\left.\frac{\partial f}{\partial \vec{l}}\right|_{(0,0)}=\lim_{\rho\to 0}\frac{f(\Delta x,\Delta y)-f(0,0)}{\rho}=\lim_{\rho\to 0}\frac{\sqrt{(\Delta x)^2+(\Delta y)^2}}{\sqrt{(\Delta x)^2+(\Delta y)^2}}=1 \text{ 存在}$$

$$\begin{cases}\left.\frac{\partial f}{\partial x}\right|_{(0,0)}=\lim\limits_{\Delta x\to 0}\frac{f(\Delta x,0)-f(0,0)}{\Delta x}=\lim\limits_{\Delta x\to 0}\frac{\sqrt{(\Delta x)^2}}{\Delta x}=\frac{|\Delta x|}{\Delta x}\\ \left.\frac{\partial f}{\partial y}\right|_{(0,0)}=\lim\limits_{\Delta y\to 0}\frac{f(0,\Delta y)-f(0,0)}{\Delta y}=\lim\limits_{\Delta y\to 0}\frac{\sqrt{(\Delta y)^2}}{\Delta y}=\frac{|\Delta y|}{\Delta y}\end{cases}$$ 偏导数不存在.

⚠函数f在(0,0)处偏导数不存在 ⇒函数f在(0,0)处不可微⇒定理 8.10 条件不满足。

补充题 2 $f(x,y)=\begin{cases}\frac{xy}{x^2+y^2}, & x^2+y^2\neq 0\\ 0, & x^2+y^2=0\end{cases}$,

在(0,0)处的某些方向的方向导数不存在，但是两个偏导数存在。

证明：函数f在(0,0)处的任何方向$\vec{l}=(\Delta x,\Delta y)$的方向导数：

$$\left.\frac{\partial f}{\partial \vec{l}}\right|_{(0,0)}=\lim_{\rho\to 0}\frac{f(\Delta x,\Delta y)-f(0,0)}{\rho}=\lim_{\rho\to 0}\frac{\frac{\Delta x\Delta y}{\rho^2}}{\rho}=\lim_{\rho\to 0}\frac{\Delta x}{\rho}\cdot\frac{\Delta y}{\rho}\cdot\frac{1}{\rho}$$ 不存在.

$$\begin{cases}\left.\frac{\partial f}{\partial x}\right|_{(0,0)}=\lim\limits_{\Delta x\to 0}\frac{f(\Delta x,0)-f(0,0)}{\Delta x}=\lim\limits_{\Delta x\to 0}\frac{0}{\Delta x}=0\\ \left.\frac{\partial f}{\partial y}\right|_{(0,0)}=\lim\limits_{\Delta y\to 0}\frac{f(0,\Delta y)-f(0,0)}{\Delta y}=\lim\limits_{\Delta y\to 0}\frac{0}{\Delta y}=0\end{cases}$$ 偏导数存在.

结论：

(3) 偏导数存在⇏方向导数存在

可微 ⇒

梯度：

$$\frac{\partial u}{\partial \vec{l}}=\frac{\partial u}{\partial x}\cdot\cos\alpha+\frac{\partial u}{\partial y}\cos\beta+\frac{\partial u}{\partial z}\cos\gamma$$

$$=\left(\frac{\partial u}{\partial x},\frac{\partial u}{\partial y},\frac{\partial u}{\partial z}\right)(\cos\alpha,\cos\beta,\cos\gamma)$$

当$\left(\frac{\partial u}{\partial x},\frac{\partial u}{\partial y},\frac{\partial u}{\partial z}\right)$与$\vec{l}$垂直 $\frac{\partial u}{\partial \vec{l}}=0$

P102 定义 8.10 $\frac{\partial u}{\partial x}\vec{i}+\frac{\partial u}{\partial y}\vec{j}+\frac{\partial u}{\partial z}\vec{k}=\left(\frac{\partial u}{\partial x},\frac{\partial u}{\partial y},\frac{\partial u}{\partial z}\right)$

称$u(P)$在P处梯度.

记为 grad u.

$$\text{grad } u \triangleq \frac{\partial u}{\partial x}\vec{i}+\frac{\partial u}{\partial y}\vec{j}+\frac{\partial u}{\partial z}\vec{k}=\left(\frac{\partial u}{\partial x},\frac{\partial u}{\partial y},\frac{\partial u}{\partial z}\right)$$

$$\text{grad}\,u|_{P_0}=\left(\frac{\partial u}{\partial x},\frac{\partial u}{\partial y},\frac{\partial u}{\partial z}\right)\Big|_{P_0}$$

$$|\text{grad}\,u|_{P_0}=\sqrt{\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial u}{\partial y}\right)^2+\left(\frac{\partial u}{\partial z}\right)^2}$$

$$\left.\frac{\partial u}{\partial l}\right|_{P_0}=\text{grad}\,u(P_0)\cdot\vec{l}_0$$

$$=|\text{grad}\,u(P_0)|\cos\theta=(\text{grad}\,u(P_0))_{l_0}$$

(1) u在点P处沿方向$\vec{l}$的方向导数 = 梯度在方向$\vec{l}$上的投影

$$\left.\frac{\partial u}{\partial \vec{l}}\right|_{P_0}=\text{gradu}(P_0)\vec{l}^0=(\text{gradu}(P_0))_{\vec{l}^0}$$

(2)当$\theta=0$，即$\vec{l}^0$的方向与梯度方向gradu(P_0)一致，取到最大值，等于梯度的模|gradu(P_0)|，即$\max\left(\left.\frac{\partial u}{\partial \vec{l}}\right|_{P_0}\right)=|\text{gradu}(P_0)|$.

当u在P_0可微时，u在P_0的梯度方向是u值增长最快的方向.

(3)当$\theta=\pi$，即$\vec{l}^0$的方向与梯度方向gradu(P_0)相反，方向导数取得最小值为$-|\text{gradu}(P_0)|$.

(4)当$\theta=\frac{\pi}{2}$，$\vec{l}^0$的方向与梯度方向 gradu(P_0)垂直，方向导数为 0.

梯度运算法则

u,v可微 α,β为常数

(1) $\text{grad}(\alpha u+\beta v)=\alpha\,\text{grad}\,u+\beta\,\text{grad}\,v$

(2) $\text{grad}(uv)=v\,\text{grad}\,u+u\,\text{grad}\,v$

(3) $\text{grad}\,f(u)=f'(u)\,\text{grad}\,u$

证明：(3)$\text{grad}f(u)=\left(\frac{\partial f(u)}{\partial x},\frac{\partial f(u)}{\partial y},\frac{\partial f(u)}{\partial z}\right)$

$$=\left(f'(u)\frac{\partial u}{\partial x},f'(u)\frac{\partial u}{\partial y},f'(u)\frac{\partial u}{\partial z}\right)=f'(u)\left(\frac{\partial u}{\partial x},\frac{\partial u}{\partial y},\frac{\partial u}{\partial z}\right)=f'(u)\text{gradu}$$

补充题 4 确定常数λ，使在右半平面$x>0$上的向量

$A(x,y)=\{2xy(x^4+y^2)^\lambda,-x^2(x^4+y^2)^\lambda\}$为具有连续二阶偏导数的某二元函数$u(x,y)$的梯度。

解：$\text{gradu}=\left(\frac{\partial u}{\partial x},\frac{\partial u}{\partial y}\right)=(2xy(x^4+y^2)^\lambda,-x^2(x^4+y^2)^\lambda)$,

$$\Rightarrow \frac{\partial u}{\partial x}=2xy(x^4+y^2)^\lambda,\qquad \frac{\partial u}{\partial y}=-x^2(x^4+y^2)^\lambda$$

$$\frac{\partial^2 u}{\partial x\partial y}=2x(x^4+y^2)^\lambda+2xy\lambda(x^4+y^2)^{\lambda-1}\cdot(2y)\quad (*)$$

$$\frac{\partial^2 u}{\partial y\partial x}=-2x(x^4+y^2)^\lambda-x^2\cdot(4x^3)\lambda(x^4+y^2)^{\lambda-1}\ (**)$$

函数$u(x,y)$具有连续二阶偏导数 $\Rightarrow \frac{\partial^2 u}{\partial x\partial y}=\frac{\partial^2 u}{\partial y\partial x}$

$(*)=(**)$得：$4x(x^4+y^2)^\lambda\cdot(\lambda+1)=0\Rightarrow\lambda=-1$.

补充题 5 求地球的万有引力势能$E=-G\frac{mM}{r}$的梯度，其中$r=\sqrt{x^2+y^2+z^2}$为矢径$\vec{r}=x\vec{i}+y\vec{j}+z\vec{k}$的模.

解：$E=-G\frac{mM}{r}=-G\frac{mM}{\sqrt{x^2+y^2+z^2}}$

$$\frac{\partial E}{\partial x}=G\frac{mM}{r^2}\cdot\frac{x}{r},\qquad \frac{\partial E}{\partial y}=G\frac{mM}{r^2}\cdot\frac{y}{r},\qquad \frac{\partial E}{\partial z}=G\frac{mM}{r^2}\cdot\frac{z}{r}$$

$$\Rightarrow \text{grad}E=\frac{\partial E}{\partial x}\vec{i}+\frac{\partial E}{\partial y}\vec{j}+\frac{\partial E}{\partial z}\vec{k}=G\frac{mM}{r^2}\left(\frac{x\vec{i}+y\vec{j}+z\vec{k}}{r}\right)=G\frac{mM}{r^2}\vec{r}^0$$

注意：万有引力势能场的梯度场就是万有引力场，沿地球的万有引力方向其引力势能的变动最快。

补充题 $f(x,y)=\begin{cases}\frac{x^2y}{x^2+y^2}, & (x^2+y^2\neq 0)\\ 0 & (x^2+y^2=0)\end{cases}$ 求 f 在原点处沿$\vec{l}$方向导数

$$\frac{\partial f}{\partial l}=\lim_{\rho\to 0}\frac{f(\rho\cos\theta,\rho\sin\theta)-f(0,0)}{\rho}=\frac{\rho^3\cos^2\alpha\sin\alpha}{\rho^3}=\cos^2\alpha\sin\alpha$$

$0 \le \left|\frac{x^2 y}{x^2+y^2}\right| \le |y|$ $\lim\limits_{\substack{x\to 0\\ y\to 0}} f(x,y) = 0$.

$f_x(0,0) \lim\limits_{\Delta x\to 0} \frac{f(\Delta x,0)-f(0,0)}{\Delta x} = 0$ 同理 $f_y(0,0)=0$

不一样 收什么？不可微

$\frac{\partial f}{\partial l} = \cos^3\alpha \sin\alpha \neq 0 = [f_x'(0,0)\ f_y'(0,0)][\cos\alpha, \sin\alpha]$

$\lim\limits_{\rho\to 0} \frac{\Delta z - dz}{\rho} = \frac{\frac{(\Delta x)^2 y}{(\Delta x)^2+(\Delta y)^2}}{\rho} = \lim\limits_{\substack{\Delta x\to 0\\ \Delta y\to 0}} \frac{(\Delta x)^2 \Delta y}{\left(\sqrt{(\Delta x)^2+(\Delta y)^2}\right)^3} = \lim\limits_{\substack{\Delta x\to 0\\ y=kx}} \frac{k^2}{(\sqrt{1+k^2})^3}$

$$f(x,y) = \begin{cases} (|x|+|y|)\sin\frac{1}{x^2+y^2} & x^2+y^2\neq 0 \\ 0 & x^2+y^2=0 \end{cases}$$

$\lim\limits_{\substack{x\to 0\\ y\to 0}} (|x|+|y|)\sin\frac{1}{x^2+y^2} = 0 = f(0,0) \Rightarrow$ 在 $(0,0)$ 连续

$\frac{\partial f}{\partial x}\Big|_{(0,0)} = \lim\limits_{\Delta x\to 0} \frac{f(\Delta x,0)-f(0,0)}{\Delta x} = \lim\limits_{\Delta x\to 0} \frac{|\Delta x \sin\frac{1}{(\Delta x)^2}|}{\Delta x}$ 不存在

同理 $\frac{\partial f}{\partial y}\Big|_{(0,0)} = \lim\limits_{\Delta y\to 0} \left|\frac{\Delta y \sin\frac{1}{(\Delta y)^2}}{\Delta y}\right|$ 不存在

一阶偏导不连续 $\Rightarrow$ 不可微

$\frac{\partial f}{\partial e}\Big|_{(0,0)} = \lim\limits_{\rho\to 0} \frac{f(\rho\sin\theta, \rho\cos\theta)-f(0,0)}{\rho} = \lim\limits_{\rho\to 0} \frac{\rho(|\cos\theta|+|\sin\theta|)\sin\frac{1}{\rho^2}}{\rho}$

$\rho = \frac{1}{\sqrt{n\pi}}$ 原式 $=1$ $\rho = \frac{1}{\sqrt{2n\pi+\frac{\pi}{2}}}$ 原式 $=-1$ $\Rightarrow$ 方向导数不存在

自变量 x 不随着其他量变化 $d(dx)=0$ ✅

$z = f(x,y)$ $d(dz) = d\left(\frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy\right)$

$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)dx + \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)dy$ $= d\left(\frac{\partial z}{\partial x}dx\right) + d\left(\frac{\partial z}{\partial y}dy\right)$

$= \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)dx^2 + \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)dy\,dx + \underbrace{d(dx)\cdot\frac{\partial z}{\partial x}}_{0} + \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)dx\,dy + \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right)dy^2 = \frac{\partial^2 z}{\partial x^2}dx^2 + \frac{\partial^2 z}{\partial x\partial y}dx\,dy + \frac{\partial^2 z}{\partial y^2}dy^2 = d^2 z$

第六节 多元函数的极值及应用

f 在 $P(x_0,y_0)$ 某个邻域 $U(P_0)$ 有直到 $n+1$ 阶偏导数. 对 $U(P_0)$ 中任一点 (x_0+h, y_0+h)

存在 $\theta\in(0,1)$, 使得

$$f(x_0+h,y_0+k)=f(x_0,y_0)+\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(x_0,y_0)+\frac{1}{2!}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^2 f(x_0,y_0)+\cdots+\frac{1}{n!}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^n f(x_0,y_0)$$

$$+\frac{1}{(n+1)!}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^{n+1} f(x_0+\theta h, y_0+\theta t)$$

称二元函数 f 在 P_0 n 阶泰勒展开式

注: $R_n \triangleq \frac{1}{(n+1)!}\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)^{n+1} f(x_0+\theta h, y_0+\theta t)$, $(0<\theta<1)$

R_n 是泰勒公式的拉格朗日余项

若$\exists M>0,\forall x\in U(P_0)$, 有$\left|f_x^{(n)}(x,y)\right|\le M(\forall n\in\mathbb{N}^+)$, 则:

$|R_n|\le\frac{M}{(n+1)!}(|h|+|k|)^{n+1}\le\frac{2^{n+1}\cdot M}{(n+1)!}\rho^{n+1}\ (\rho=\sqrt{h^2+k^2})$.

表明: $R_n=o(\rho^n),(\rho\to 0)R_n$(比$\rho^n$高阶的无穷小)

(2)当$(x_0,y_0)=(0,0)$时, 则称为二元函数的麦克劳林公式。

(3) 二元函数的中值公式 (当$n=0$时)

$$f(x_0+h,y_0+k)-f(x_0,y_0)=\left(h\frac{\partial}{\partial x}+k\frac{\partial}{\partial y}\right)f(x_0+\theta h,y_0+\theta k)$$

$$=f_x'(x_0+\theta h,y_0+\theta k)h+f_y'(x_0+\theta h,y_0+\theta k)k$$

这就是二元函数的拉格朗日中值公式。

(4)推论 设$f(x,y)$在区域G上具有连续的一阶偏导.

$O(x^n)$ 皮亚诺余项

R_n 拉格朗日余项

若 $f_x'(x,y)\equiv 0$, $(x,y)\in G$, 则 $f(x,y)$ 在 G 上仅是 x 的函数

若 $f_y'(x,y)\equiv 0$, $(x,y)\in G$, 则 $f(x,y)$ 在 G 上仅是 y 的函数

若 $f_x'(x,y)\equiv 0$, $f_y'(x,y)\equiv 0$, 则 $f(x,y)$ 在 G 上是常值函数

提示:

(i)只需证 $\forall(x_1,y_0),(x_2,y_0)\in G\Rightarrow f(x_1,y_0)=f(x_2,y_0)$

设$\psi(x)\triangleq f(x,y_0),\psi'(x)=f_x'(x,y_0)$,

不妨设$x_1<x_2$,$\psi(x)$在区间$[x_1,x_2]$上利用拉格朗日定理得:

$f(x_2,y_0)-f(x_1,y_0)=\psi(x_2)-\psi(x_1)=\psi'(\xi)(x_1-x_2)$

$=f_x'(\xi,y_0)(x_1-x_2)=0\ (x_1<\xi<x_2)\Rightarrow$(i)成立

同理可证(ii)成立.

(iii)$\forall(x_1,y_1),(x_2,y_2)\in G$, 由二元函数中值定理得:

$f(x_2,y_2)-f(x_1,y_1)=f(x_1+(x_2-x_1),y_1+(y_2-y_1))-f(x_1,y_1)$

$=(x_2-x_1)f_x'(x_1+\theta(x_2-x_1),y_1+\theta(y_2-y_1))+$

$+(y_2-y_1)f_y'(x_1+\theta(x_2-x_1),y_1+\theta(y_2-y_1))=0$

$\Rightarrow f(x,y)$在G上是常值函值.

P.107 例 1 写出$f(x,y)=x^y$在$P_0(1,1)$邻域内的展开式(二次项为止)。

解: $\begin{cases}\frac{\partial f}{\partial x}=yx^{y-1}\\ \frac{\partial f}{\partial y}=x^y\ln x\end{cases}\Rightarrow\begin{cases}\frac{\partial^2 f}{\partial x^2}=y(y-1)x^{y-2}\\ \frac{\partial^2 f}{\partial x\partial y}=yx^{y-1}\ln x+x^{y-1}\\ \frac{\partial^2 f}{\partial y^2}=x^y\ln^2 x\end{cases}$

$\Rightarrow\begin{cases}\frac{\partial^3 f}{\partial x^3}=y(y-1)(y-2)x^{y-3}\\ \frac{\partial^3 f}{\partial x^2\partial y}=(2y-1)x^{y-2}+y(y-1)x^{y-2}\ln x\\ \frac{\partial^3 f}{\partial x\partial y^2}=yx^{y-1}\ln^2 x+2x^{y-1}\ln x\\ \frac{\partial^3 f}{\partial y^3}=x^y\ln^3 x\end{cases}$

$h\triangleq x-1, k\triangleq y-1$,由二元函数的泰勒公式得:

$x^y=f(x,y)=f(1+(x-1),1+(y-1))\quad(0<\theta<1)$

$=1+(x-1)+(x-1)(y-1)+R_2(1+\theta(x-1),1+\theta(y-1))$

其中: $R_2(x,y)=\frac{1}{3!}\{y(y-1)(y-2)x^{y-3}h^3+3[(2y-1)x^{y-2}+y(y-1)x^{y-2}\ln x]h^2k$

$+3[yx^{y-1}\ln^2 x+2x^{y-1}\ln x]hk^2+x^y\ln^3 x k^3\}$

注: 余项$R_2(1+\theta(x-1),1+\theta(y-1))$的形式很复杂, 故若不需要指出, 一般写成$R_n$或$o(\rho^n)$, 如书本该题结果为:

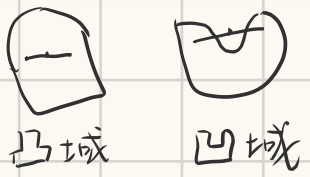
$x^y=1+(x-1)+(x-1)(y-1)+o(\rho^2)\ (\rho\triangleq\sqrt{h^2+k^2}\to 0)$

二元函数中值定理

凸域：$P_1(x_1,y_1)$ $P_2(x_2,y_2)$

$P_1, P_2 \in D$ $\forall \lambda \in [0,1]$

若 $P[x_1+\lambda(x_2-x_1), y_1+\lambda(y_2-y_1)] \in D$ 则为凸域



设函数 $f(x,y)$ 在凸域 D 上连续，在 D 内所有点均可微，

则 对 D 内任意两点 $P(x_0,y_0)$ $Q(x_0+h, y_0+t)$

$$f(x_0+h, y_0+k) - f(x_0,y_0) = f_x'(x_0+\theta h, y_0+\theta k)h + f_y'(x_0+\theta h, y_0+\theta k)k$$

证明大致思路：

令$\varphi(t) \triangleq f(x_0+th, y_0+tk)$，则$\varphi(t)$在$[0,1]$上连续在$(0,1)$内可导，

且$\varphi'(t) = f_x'(x_0+th, y_0+tk)h + f_y'(x_0+th, y_0+tk)k$

由一元函数的*Lagrange*中值定理得：$\exists\theta \in (0,1)$使得：

$\varphi(1)-\varphi(0) = \varphi'(\theta) = f_x'(x_0+\theta h, y_0+\theta k)h + f_y'(x_0+\theta h, y_0+\theta k)k$

因为D是凸域，则点$(x_0+\theta h, y_0+\theta k) \in D$.

以t为变量的 $\varphi(t)$

将二元函数化为一元

计算 $\lim\limits_{\substack{x\to 0\\ y\to 0}} \dfrac{\ln(x^2+e^{y^2})}{x^2+y^2}$

提示：法一 $f(x,y) \triangleq \ln(x^2+e^{y^2}) \Rightarrow \begin{cases} f_x'(x,y) = \dfrac{2x}{x^2+e^{y^2}} \\ f_y'(x,y) = \dfrac{2ye^{y^2}}{x^2+e^{y^2}} \end{cases}$

$$\Rightarrow \begin{cases} f_{xx}''(x,y) = \dfrac{2e^{y^2}-2x^2}{(x^2+e^{y^2})^2} \\ f_{xy}''(x,y) = \dfrac{-4xye^{y^2}}{(x^2+e^{y^2})^2} \\ f_{yy}''(x,y) = \dfrac{(2+4y^2)e^{y^2}(x^2+e^{y^2}) - 4y^2e^{y^2}}{(x^2+e^{y^2})^2} \end{cases} \Rightarrow \begin{cases} f_x'(0,0) = f_y'(0,0) = 0 \\ f_{xy}''(0,0) = 0 \\ f_{yx}''(0,0) = f_{yy}''(0,0) = 2 \end{cases}$$

由*Taylor*公式得：$f(0+x, 0+y) = f(0,0) + xf_x' + yf_y' + \frac{1}{2!}(f_{xx}''x^2 + 2xyf_{xy}'' + y^2f_{yy}'') = x^2+y^2$

$$f(x,y) = (x^2+y^2) + o(\rho^2) \quad (\rho \triangleq \sqrt{x^2+y^2} \to 0) \Rightarrow \lim_{\substack{x\to 0\\ y\to 0}} \frac{\ln(x^2+e^{y^2})}{x^2+y^2} = 1$$

法二 当$(x,y)\to(0,0)$时，有：

$$\ln(x^2+e^{y^2}) = \ln\left(1-(x^2+e^{y^2}-1)\right) \sim (x^2+e^{y^2}-1)$$

$$\Rightarrow \lim_{\substack{x\to 0\\ y\to 0}} \frac{\ln(x^2+e^{y^2})}{x^2+y^2} = \lim_{\substack{x\to 0\\ y\to 0}} \frac{x^2+e^{y^2}-1}{x^2+y^2} = 1 + \lim_{\substack{x\to 0\\ y\to 0}} \frac{e^{y^2}-1-y^2}{x^2+y^2} = 1$$

本质、弄清 h,k 代入

多元函数的极值

定理 8.12 极值必要条件

若 f 在 $P_0(x_0,y_0)$ 存在 偏导数且在 P_0 取极值，

则 $f_x'(x_0,y_0)=0$ $f_y'(x_0,y_0)=0$

满足这两个式子的点：稳定点、驻点

分析：若f在点(x_0,y_0)取极值，则一元函数$f(x,y_0)$必定在$x=x_0$取相同的极值。当$f_x'(x_0,y_0)$存在，即$\left.\dfrac{d}{dx}f(x,y_0)\right|_{x=x_0}$存在，

由费马定理知：$\left.\dfrac{d}{dx}f(x,y_0)\right|_{x=x_0} = 0$，即$f_x'(x_0,y_0)=0$.

同理一元函数$f(x_0,y)$在$y=y_0$也取相同的极值，若$f_y'(x_0,y_0)$也存在，则$f_y'(x_0,y_0)=0$.

注(1) f存在偏导数，其极值点一定是稳定点

(2)反之不一定成立，如 $f(x,y)=xy$ 在点 $O(0,0)$ 处不取极值.

$\because \begin{cases} f_x'(x,y)=y \\ f_y'(x,y)=x \end{cases} \Rightarrow \begin{cases} f_x'(0,0)=0 \\ f_y'(0,0)=0 \end{cases}$ 而 $\begin{cases} f(P)>0,\ P\text{在 I, III 象限} \\ f(P)<0,\ P\text{在 II, IV 象限} \end{cases}$

稳定点

(3)函数 $f(x,y)$ 在点 (x_0,y_0) 取极值，$f(x,y)$ 的偏导数只有两种情形:

(i) $f_x'(x_0,y_0)$，$f_y'(x_0,y_0)$ 都存在，则 $f_x'(x_0,y_0)=0$，$f_y'(x_0,y_0)=0$.

即点 $P_0(x_0,y_0)$ 为稳定点.

(ii) $f_x'(x_0,y_0)$，$f_y'(x_0,y_0)$ 至少有一个不存在.

不稳定

因此，$f(x,y)$ 的极值点一定包含在稳定点或偏导数不存在点之中.

如: P.108 例 2 设 $f(x,y)=\sqrt{x^2+y^2}$，$f(x,y)$ 在点 $(0,0)$ 处偏导数不存在，

$\forall (x,y)\in R$ 有 $f(x,y)\geq f(0,0)=0 \Rightarrow f(0,0)$ 为极小值.

极值充分条件

$z=f(x,y)$ 在定 (x_0,y_0) 某邻域 $U(P_0)$ 连续.

且有一阶二阶偏导数 $f_x'(x_0,y_0)=0$ $f_y'(x_0,y_0)=0$

设 $A=f_{xx}''(x_0,y_0)$，$B=f_{xy}''(x_0,y_0)$ $C=f_{yy}''(x_0,y_0)$

(1) $B^2-AC<0$，$f(x_0,y_0)$ 一定极值

A (或 C) >0 $f(x_0,y_0)$ 极小值

A (或 C) <0 $f(x_0,y_0)$ 极大值

(2) $B^2-AC>0$，$f(x_0,y_0)$ 不是极值

(3) $B^2-AC=0$ 不能判断 $f(x_0,y_0)$ 是否极值，进一步研究

P.110 例 3 求函数 $f(x,y)=x^3+y^3-3xy$ 的极值. 注: 留意该题解题步骤

解: $\begin{cases} \frac{\partial f}{\partial x}=3x^2-3y=0 \\ \frac{\partial f}{\partial y}=3y^2-3x=0 \end{cases} \Rightarrow \begin{cases} x=0 \\ y=0 \end{cases}, \begin{cases} x=1 \\ y=1 \end{cases} \Rightarrow P_0(0,0), P_1(1,1).$

函数无偏导数不存在的点.

$\begin{cases} \frac{\partial^2 f}{\partial x^2}=6x \\ \frac{\partial^2 f}{\partial x\partial y}=-3 \\ \frac{\partial^2 f}{\partial y^2}=6y \end{cases} \Rightarrow$ P_0: $\begin{cases} A=0 \\ B=-3 \\ C=0 \\ B^2-AC>0 \end{cases}$ P_1: $\begin{cases} A=6 \\ B=-3 \\ C=6 \\ B^2-AC<0, A>0 \end{cases}$

$\Downarrow$ $\Downarrow$

$f(0,0)$ 不是极值 $f(1,1)=-1$ 是极小值

(或列表如下，两者取其一)

	A	B	C	B^2-AC	
$P_0(0,0)$	0	−3	0	$9>0$	$f(0,0)=0$ 非极值
$P_1(1,1)$	6	−3	6	$-27<0$	$f(1,1)=-1$ 极小值

补充题2 讨论函数$f(x,y)=(y-x^2)(y-2x^2)$在原点是否有极值?

解: $\begin{cases}\dfrac{\partial f}{\partial x}=8x^3-6xy=0\\ \dfrac{\partial f}{\partial y}=-3x^2+2y=0\end{cases}\Rightarrow\begin{cases}x=0\\ y=0\end{cases}$ $P(0,0)$ $\begin{cases}\dfrac{\partial^2 f}{\partial x^2}=24x^2-6y\\ \dfrac{\partial^2 f}{\partial x\partial y}=-6x\\ \dfrac{\partial^2 f}{\partial y^2}=2\end{cases}\Rightarrow^{P}\begin{cases}A=0\\ B=0\\ C=2\\ B^2-AC=0\end{cases}$

$\begin{cases}f(x,y)>0=f(0,0), & y>2x^2\\ f(x,y)<0=f(0,0), & x^2<y<2x^2\end{cases}\Rightarrow f(0,0)$不是极值.

注: $f(x,y)=(y-x^2)(y-2x^2)=2x^4-3x^2y+y^2$

二、多元函数的最大值、最小值

$f(P)$ 在闭区域 G 连续

在 G 上一定取到最大(小)值

$\exists P_1, P_2\in G$, $f(P_1)=m$, $f(P_2)=M$

$\forall P\in G$ $m\le f(P)\le M$.

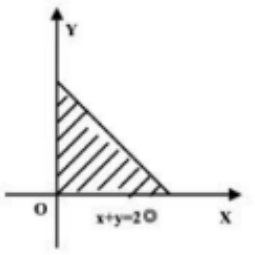
最值在 { 边界值 / 区域内部. } 取到 (最大, 最小) 一定是极值点 { 稳定点 / 偏导数不存在点 }

一元函数 定义域 边界—两端点

闭区域 G 边界 ⟶ 无数个点 { $G\subset R^2$ 边界 ↔ 边界 曲线 / $G\subset R^3$ 边界 ↔ 边界 曲面 }

(2) 若实际问题一定有最大值(或最小值),且内部有唯一的可疑点,则该点的函数无须判断一定是最大值(或最小值).

P.111 例 4 设D是由x轴,y轴及直线$x+y=2\pi$所围成的三角形区域.求函数$u=\sin x+\sin y-\sin(x+y)$在$D$上的最大值.



解 函数没有偏导数不存在的点,偏导数为零的点:

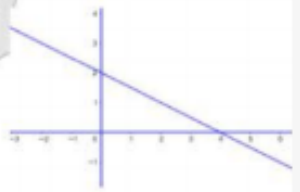
$\begin{cases}\dfrac{\partial u}{\partial x}=\cos x-\cos(x+y)=0\\ \dfrac{\partial u}{\partial y}=\cos y-\cos(x+y)=0\end{cases}\Rightarrow\begin{cases}x=\dfrac{2}{3}\pi\\ y=\dfrac{2}{3}\pi\end{cases},\begin{cases}x=0\\ y=0\end{cases},\begin{cases}x=2\pi\\ y=0\end{cases},\begin{cases}x=0\\ y=2\pi\end{cases}$

$\Rightarrow u\left(\dfrac{2}{3}\pi,\dfrac{2}{3}\pi\right)=\dfrac{3\sqrt{3}}{2}, u(0,0)=u(0,2\pi)=u(2\pi,0)=0.$

在边界$x=0$, $y=0$, $x+y=2\pi\Rightarrow u(x,y)=0.$

$\left(\dfrac{2}{3}\pi,\dfrac{2}{3}\pi\right)$三角形区域内部唯一可疑点 $\Rightarrow u\left(\dfrac{2}{3}\pi,\dfrac{2}{3}\pi\right)=\dfrac{3\sqrt{3}}{2}>0$ 为最大值.

补充题 3 求平面上一点$P(x,y)$使得其到直线$x+2y-4=0$及Ox轴、Oy轴的距离平方和最小。



解： 令$P(x,y)$使得其到直线及Ox轴、Oy轴的距离平方和为$f(x,y)$:

$$f(x,y)\triangleq\frac{1}{5}(x+2y-4)^2+x^2+y^2$$

区域内部:

$$\begin{cases}\dfrac{\partial u}{\partial x}=\dfrac{2}{5}(x+2y-4)+2x=0\\ \dfrac{\partial u}{\partial y}=\dfrac{4}{5}(x+2y-4)+2y=0\end{cases}\Rightarrow\begin{cases}x=\dfrac{2}{5}\\ y=\dfrac{4}{5}\end{cases}\left(\begin{matrix}\text{唯一}\\ \text{驻点}\end{matrix}\right),\begin{cases}\dfrac{\partial^2u}{\partial x^2}=\dfrac{12}{5}=A\\ \dfrac{\partial^2u}{\partial x\partial y}=\dfrac{4}{5}=B\\ \dfrac{\partial^2u}{\partial y^2}=\dfrac{18}{5}=C\end{cases}$$

$\Rightarrow$ 驻点$P\left(\frac{2}{5},\frac{4}{5}\right),B^2-AC<0,A>0\Rightarrow f\left(\frac{2}{5},\frac{4}{5}\right)=\frac{8}{5}$极小值

边界上：三条边界直线的交点：$f(0,0)=\frac{16}{5},f(0,2)=4,f(4,0)=16.$

边界 1　$x+2y-4=0\ (0\le y\le 2)\Rightarrow x=2(2-y)$

$$\Rightarrow f(x,y)=5\left(y-\frac{8}{5}\right)^2+16-5\cdot\frac{64}{25}\Rightarrow\left(f(x,y)|_{\substack{x+2y-4=0\\0\le y\le 2}}\right)_{MIN}=f\left(\frac{4}{5},\frac{8}{5}\right)=\frac{16}{5}$$

边界 2　Oy轴:　$x=0\ (0\le y\le 2)$

$$\Rightarrow f(0,y)=\frac{1}{5}\left[9\left(y-\frac{8}{9}\right)^2+\frac{80}{9}\right]\Rightarrow\left(f(x,y)|_{\substack{x=0\\0\le y\le 2}}\right)_{MIN}=f\left(0,\frac{8}{9}\right)=\frac{16}{9}$$

边界 3　Ox轴:　$y=0\ (0\le x\le 4)$

$$\Rightarrow f(x,0)=\frac{2}{5}\left[3\left(y-\frac{2}{3}\right)^2+\frac{20}{3}\right]\Rightarrow\left(f(x,y)|_{\substack{y=0\\0\le x\le 4}}\right)_{MIN}=f\left(0,\frac{2}{3}\right)=\frac{8}{3}$$

综上：$\frac{8}{5}<\frac{16}{9}<\frac{8}{3}<\frac{16}{5}<4<16$

则：平面上点$P\left(\frac{2}{5},\frac{4}{5}\right)$到直线$x+2y-4=0$及$Ox$轴、$Oy$轴的距离平方和最小。

三、条件极值

在一定条件约束下的极值

条件极值问题 —消元法→ 无条件极值

例如： 要设计一个容量为V的长方体无上盖水箱，试问水箱长，宽，高各等于多少时，其所用的材料最少（即表面积最小）.

分析： 设水箱的长、宽、高分别为x,y,z，则表面积为

$$S(x,y,z)=2(xz+yz)+xy\qquad(1)$$

而目标函数$s(x,y,z)$的定义域是$x>0,y>0,z>0$，且需满足条件:

$$xyz=v\qquad(2)$$

像这类附有约束条件的极值问题称为条件极值问题，

不带约束条件的极值问题称为无条件极值问题.

方法一： 条件极值问题 $\overset{\text{消元法}}{\Rightarrow}$ 无条件极值问题

$xyz=v\Rightarrow z=\frac{v}{xy}$，代入$S(x,y,z)=2(xz+yz)+xy$ 得:

$$F(x,y)\triangleq s\left(x,y,\frac{v}{xy}\right)=2v\left(\frac{1}{y}+\frac{1}{x}\right)+xy\ (x>0,y>0),$$

$F(x,y)$在定义域内无偏导数不存在的点,解方程组

$$\begin{cases}\dfrac{\partial F}{\partial x}=-2v\dfrac{1}{x^2}+y=0\\ \dfrac{\partial F}{\partial y}=-2v\dfrac{1}{y^2}+x=0\end{cases}\Rightarrow\begin{cases}x=\sqrt[3]{2v}\\ y=\sqrt[3]{2v}\\ z=\dfrac{1}{2}\sqrt[3]{2v}\end{cases}\text{唯一驻点}$$

由实际问题表面积无最大值，只有最小值，则:

当$x=y=\sqrt[3]{2v},z=\frac{1}{2}\sqrt[3]{2v}$时,表面积$s=3\sqrt[3]{4v^2}$最小。

方法二：拉格朗日乘数法

拉格朗日函数 $L(x,y,z,\lambda)=2(xz+yz)+xy+\lambda(xyz-v)$

$$\begin{cases}\dfrac{\partial L}{\partial x}=2z+y+\lambda yz=0\\ \dfrac{\partial L}{\partial y}=2z+x+\lambda zx=0\\ \dfrac{\partial L}{\partial z}=2(x+y)+\lambda xy=0\\ \dfrac{\partial L}{\partial \lambda}=xyz-v=0\end{cases}\Rightarrow\begin{cases}x=y=2z=\sqrt[3]{2v}\\ \lambda=-\dfrac{4}{\sqrt[3]{2v}}\end{cases}\text{唯一驻点}$$

根据题意，所求水箱的表面积在条件$xyz=v$下确实存在最小值,

所以当高为$z=\sqrt[3]{\frac{v}{4}}$，长x与宽y为高的 2 倍时表面积最小，最小值$S=3\sqrt[3]{4v^2}$.

注意： 一定要根据实际问题判别是否为最值点（或极值点）。

三、条件极值（拉格朗日乘子法）

$\max(\min)\ u = f(x,y,z)$ （目标函数）

$s.t.\ \ G(x,y,z)=0$ （约束函数）

f, G 具有连续偏导且 $\frac{\partial G}{\partial z}\neq 0$ 隐函数存在定理

$G(x,y,z)=0$ 确定隐函数 $z=z(x,y)\left(\frac{\partial z}{\partial x}=-\frac{G_x'}{G_z'},\ \frac{\partial z}{\partial y}=-\frac{G_y'}{G_z'}\right)$.

处理思路：消元法 $G(x,y,z)=0$ 解出 $z=z(x,y)$ 代入目标函数 f：

条件极值问题		无条件极值问题
$u=f(x,y,z)$ 极值	$\Rightarrow$	$u=f(x,y,z(x,y))$ 极值
$s.t.\ \ G(x,y,z)=0$		

设 (x_0,y_0) 为 $u=f(x,y,z(x,y))$ 极值点 $(z_0=z(x_0,y_0))$，满足二元函数的极值的必要条件（P.108 定理 8.12）：

$$\begin{cases}\frac{\partial u}{\partial x}=0,\\ \frac{\partial u}{\partial y}=0.\end{cases}\xrightarrow[\text{求导}]{\text{复合函数}\Rightarrow}\begin{cases}\frac{\partial u}{\partial x}=f_x'+f_z'\frac{\partial z}{\partial x}=f_x'-\frac{G_x'}{G_z'}f_z'=0,\\ \frac{\partial u}{\partial y}=f_y'+f_z'\frac{\partial z}{\partial y}=f_y'-\frac{G_y'}{G_z'}f_z'=0.\end{cases}\quad z=z(x,y)$$

$\Leftrightarrow$ 所求问题的解 (x_0,y_0,z_0) 必须满足关系式：

$$\frac{f_x'(x_0,y_0,z_0)}{G_x'(x_0,y_0,z_0)}=\frac{f_y'(x_0,y_0,z_0)}{G_y'(x_0,y_0,z_0)}=\frac{f_z'(x_0,y_0,z_0)}{G_z'(x_0,y_0,z_0)}\triangleq -\lambda.$$

综上所述，(x_0,y_0,z_0) 是以下方程组的解：

$$\begin{cases}f_x'(x,y,z)+\lambda G_x'(x,y,z)=0\\ f_y'(x,y,z)+\lambda G_y'(x,y,z)=0\\ f_z'(x,y,z)+\lambda G_z'(x,y,z)=0\\ G(x,y,z)=0\end{cases}\begin{matrix}(*)\\ (**)\\ (\text{约束函数})\end{matrix}$$

另一方面：若函数 $L(x,y,z,\lambda)\triangleq f(x,y,z)+\lambda G(x,y,z)$ 存在极值点 (x^*,y^*,z^*,λ^*)，则满足极值的必要条件，

$$\begin{cases}L_x'(x,y,z,\lambda)=f_x'(x,y,z)+\lambda G_x'(x,y,z)=0\\ L_y'(x,y,z,\lambda)=f_y'(x,y,z)+\lambda G_y'(x,y,z)=0\\ L_z'(x,y,z,\lambda)=f_z'(x,y,z)+\lambda G_z'(x,y,z)=0\\ L_\lambda'(x,y,z,\lambda)=G(x,y,z)=0\end{cases}$$

$\Rightarrow (x^*,y^*,z^*,\lambda^*)$ 也是方程组 (**) 的解 $\Rightarrow (x^*,y^*,z^*)=(x_0,y_0,z_0)$

进一步：推广到多变量与多约束条件的情形：一般条件极值问题

$$\max(\min)\ y=f(x_1,x_2,\cdots,x_n)$$
$$s.t.\ \ \psi_k(x_1,x_2,\cdots,x_n)=0, k=1,2,\ldots m(m<n)$$

注意：约束条件应该等式约束：$\psi_k(x_1,x_2,\cdots,x_n)=0$

此时拉格朗日函数是：

$$L(x_1,x_2,\cdots x_n,\lambda_1,\lambda_2,\cdots\lambda_m)=f(x_1,x_2,\cdots x_n)+\sum_{k=1}^{m}\lambda_k\psi_k(x_1,x_2,\cdots x_n)$$

其中 $\lambda_1,\lambda_2,\cdots\lambda_m$ 为拉格朗日乘数.

⚠最后一步：鉴于物理意义、实际经验来判断是否为极值点！

注意：约束条件应该写成的形式：$\psi_k(x_1,x_2,\cdots,x_n)=0$

P.114 例 6 求内接于椭球面 $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ 的体积最大的长方体.

（内接于椭球面的长方体的体积没有最小值，存在最大值）

提示：设该内接长方体的体积为 V，$P(x,y,z).(x>0,y>0,z>0)$ 是长方体的一个顶点且位于椭球面上即

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1,\ V=8xyz\ (x>0,y>0,z>0)$$

转化为条件极值问题：$\max V=8xyz$，$s.t.\ \frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1\ (x>0,y>0,z>0)$

$$L(x,y,z,\lambda)=8xyz+\lambda\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1\right)$$

$$\begin{cases}\frac{\partial L}{\partial x}=8yz+\frac{2\lambda x}{a^2}=0 & (1)\\ \frac{\partial L}{\partial y}=8xz+\frac{2\lambda y}{b^2}=0 & (2)\\ \frac{\partial L}{\partial z}=8xz+\frac{2\lambda z}{c^2}=0 & (3)\\ \frac{\partial L}{\partial \lambda}=\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1=0 & (4)\end{cases}\xrightarrow{x\cdot(1)+y\cdot(2)+z\cdot(3)}\begin{cases}8xyz=-\frac{2\lambda x^2}{a^2}\\ 8xyz=-\frac{2\lambda y^2}{b^2}\\ 8xyz=-\frac{2\lambda z^2}{c^2}\end{cases}$$

$$\Rightarrow\frac{2\lambda x^2}{a^2}=\frac{2\lambda y^2}{b^2}=\frac{2\lambda z^2}{c^2}\Rightarrow\frac{x^2}{a^2}=\frac{y^2}{b^2}=\frac{z^2}{c^2}\ \text{或}\lambda=0\ (8xyz=0,\ \text{不合题意舍})$$

$$\frac{x^2}{a^2}=\frac{y^2}{b^2}=\frac{z^2}{c^2}\xrightarrow[(4)]{\text{代入}}\frac{3z^2}{c^2}-1=0\Rightarrow z=\frac{c}{\sqrt{3}}\Rightarrow\begin{cases}x=\frac{a}{\sqrt{3}}\\ y=\frac{b}{\sqrt{3}}\end{cases}\ \text{唯一驻点}\Rightarrow\left(\frac{a}{\sqrt{3}},\frac{b}{\sqrt{3}},\frac{c}{\sqrt{3}}\right)$$

为顶点且对称于坐标平面的长方体为所求的最大体积长方体，最大体积为：

$$V=\frac{8abc}{3\sqrt{3}}.$$

注：λ 是辅助参数，不一定要求解，只需求出 (x_0,y_0,z_0)。

即每个条件主导数求导后为0代入点

条件极值实际应用：几何优化问题$\Rightarrow$条件极值问题

例如 1：点到平面的距离公式推导：

求点 $P_0(x_0,y_0,z_0)$ 到平面 $\pi: Ax+By+Cz+D=0$ 的距离。

解：转化为条件极值问题：$\min\ (x-x_0)^2+(y-y_0)^2+(z-z_0)^2$，$s.t.\ \ Ax+By+Cz+D=0$

$$L(x,y,z,\lambda)=[(x-x_0)^2+(y-y_0)^2+(z-z_0)^2]+2\lambda(Ax+By+Cz+D)$$

$$\begin{cases}L_x'=2(x-x_0)+2\lambda A=0\\ L_y'=2(y-y_0)+2\lambda B=0\\ L_z'=2(z-z_0)+2\lambda C=0\\ L_\lambda'=2(Ax+By+Cz+D)=0\end{cases}\Rightarrow\begin{cases}x=x_0-\lambda A\\ y=y_0-\lambda B\\ z=z_0-\lambda C\\ \lambda=\frac{Ax_0+By_0+Cz_0+D}{A^2+B^2+C^2}\end{cases}\ \text{唯一驻点}$$

事实上点 P_0 到平面 π 的最小距离一定存在，驻点 $P(x_0-\lambda A,y_0-\lambda B,z_0-\lambda C)$，$|P_0P|$ = 点 P_0 到平面 π 的距离：

$$d=|P_0P|=|\lambda_0|\sqrt{A^2+B^2+C^2}=\frac{|Ax_0+By_0+Cz_0+D|}{\sqrt{A^2+B^2+C^2}}$$

P.116 例 7 求$z=x^2+y^2+2x+y$在区域$x^2+y^2\le 1$上的最大值与最小

提示 $\because x^2+y^2\le 1$是有界闭区域，$z=x^2+y^2+2x+y$

在该区域上连续 $\Rightarrow$ 一定在该区域存在最大值与最小值.

$$(1)\begin{cases}\dfrac{\partial z}{\partial x}=2x+2=0\\ \dfrac{\partial z}{\partial y}=2y+1=0\end{cases}\Rightarrow\begin{cases}x=-1\\ y=-\dfrac{1}{2}\end{cases}\left(\begin{array}{l}\text{舍}\because (-1)^2+\left(-\dfrac{1}{2}\right)^2>1\\ \Rightarrow\left(-1,-\dfrac{1}{2}\right)\text{不在区域内}\end{array}\right)$$

(2) 函数在区域内部无偏导数不存在的点.

(3)边界上$(x^2+y^2=1\Rightarrow z=1+2x+y)$的最值点

即求条件极值问题: $\max\ (\min)\ z=1+2x+y$
$s.t.\ x^2+y^2=1$

作拉格朗日函数: $L(x,y,\lambda)=1+2x+y+\lambda(x^2+y^2-1)$,

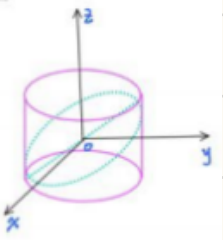
$$\begin{cases}\dfrac{\partial L}{\partial x}=2\lambda x+2=0 & (1)\\ \dfrac{\partial L}{\partial y}=2\lambda y+1=0 & (2)\\ \dfrac{\partial L}{\partial \lambda}=x^2+y^2-1=0 & (3)\end{cases}\xrightarrow[\text{代入(3)}]{(1),(2)\ \Rightarrow}\begin{cases}x=-\dfrac{1}{\lambda}\\ y=-\dfrac{1}{2\lambda}\\ \dfrac{1}{\lambda^2}+\dfrac{1}{4\lambda^4}-1=0\end{cases}$$

$$\Rightarrow\begin{cases}x=\dfrac{2}{\sqrt5}\\ y=\dfrac{1}{\sqrt5}\\ \lambda=-\dfrac{\sqrt5}{2}\end{cases},\begin{cases}x=-\dfrac{2}{\sqrt5}\\ y=-\dfrac{1}{\sqrt5}\\ \lambda=\dfrac{\sqrt5}{2}\end{cases}\Rightarrow\begin{array}{l}z\left(\dfrac{2}{\sqrt5},\dfrac{1}{\sqrt5}\right)=1+\sqrt5\\ z\left(-\dfrac{2}{\sqrt5},-\dfrac{1}{\sqrt5}\right)=1-\sqrt5\end{array}\left(\begin{array}{l}\text{怀疑点}\\ \text{仅两个}\end{array}\right)$$

$\Rightarrow$最小值$m=1-\sqrt5$, 最大值$M=1+\sqrt5$.

例如 2 平面截立体所得的截面面积

计算平面$\pi: x+y+z=0$截立体$x^2+y^2\le 1$所得的截面面积S。



解: 相交曲线$L:\begin{cases}x^2+y^2=1\\ x+y+z=0\end{cases}$(椭圆),其面积为$S=\pi ab$

$\Leftarrow$ 椭圆L上点到原点距离d最大值(a)和最小值(b)

转化为条件极值问题:
$$\begin{array}{ll}\min & f(x,y,z)\triangleq d^2=x^2+y^2+z^2\\ s.t. & x+y+z=0\\ & x^2+y^2=1\end{array}$$

$L(x,y,z,\lambda)=(x^2+y^2+z^2)+2\lambda(x+y+z)+\mu(x^2+y^2-1)$

$$\begin{cases}L'_x=2x+2\lambda+2\mu x=0 & (1)\\ L'_y=2y+2\lambda+2\mu y=0 & (2)\\ L'_z=2z+2\mu=0\\ L'_\lambda=2(x+y+z)=0\\ L'_\mu=x^2+y^2-1=0\end{cases}\xrightarrow{(1)-(2)}(1+\mu)(x-y)=0$$

$$\begin{array}{l}\mu=-1\\ \lambda=0\end{array}\Rightarrow\begin{cases}x=\pm\dfrac{1}{\sqrt2}\\ y=\mp\dfrac{1}{\sqrt2}\\ z=0\end{cases}\Rightarrow d=\sqrt{x^2+y^2+z^2}=1,$$

$$\begin{array}{l}\mu\ne -1\\ x=y\end{array}\Rightarrow\begin{cases}x=\pm\dfrac{1}{\sqrt2}\\ y=\pm\dfrac{1}{\sqrt2}\\ z=\mp\sqrt2\end{cases}\Rightarrow d=\sqrt{x^2+y^2+z^2}=\sqrt3.$$

$\Rightarrow a=\sqrt3, b=1\Rightarrow S=\pi ab=\sqrt3\pi$

思考题: 抛物面$z=x^2+y^2(\ge 0)$被平面$x+y+z=1$截成一个椭圆，求该椭圆上点到原点距离的最大值和最小值。截面椭圆的面积S为多少?

分析: 与上题类似: $\begin{cases}d_{\max}=\text{椭圆上点到原点距离的最大值}\\ d_{\min}=\text{椭圆上点到原点距离的最小值}\end{cases}$

转化为条件极值问题:
$$\begin{array}{ll}\min & f(x,y,z)\triangleq d^2=x^2+y^2+z^2\\ s.t. & x+y+z=1\\ & z=x^2+y^2\Leftrightarrow (x^2+y^2-z=0)\end{array}$$

$L(x,y,z,\lambda)=(x^2+y^2+z^2)+2\lambda(x+y+z-1)+\mu(x^2+y^2-z)$

$$\Rightarrow\begin{cases}d_{\max}=\sqrt{9+5\sqrt3}\\ d_{\min}=\sqrt{9-5\sqrt3}\end{cases}$$

计算截面椭圆的面积S，利用投影柱面 ——消去z

$$D:\begin{cases}x+y+x^2+y^2=1\\ z=0\end{cases}\Leftrightarrow\begin{cases}\left(x-\dfrac{1}{2}\right)^2+\left(y-\dfrac{1}{2}\right)^2=\dfrac{3}{2}\\ z=0\end{cases}\Rightarrow S_D=\frac{3}{2}\pi.$$

平面π法向量为$\vec n=(1,1,1)$与平面Oxy法矢量$(0,0,1)$的夹角为θ,

$$\Rightarrow S=\frac{S_D}{|\cos\theta|}=\frac{3\sqrt3}{2}\pi.$$

第七节 偏导数在几何上的应用

矢值函数 $P\ (x(t), y(t), z(t))$

$$\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$$

$$\Delta\vec{r} = \vec{r}(t+\Delta t) - r(t)$$

$$\vec{r}'(t) = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t} = x'(t)\vec{i} + y'(t)\vec{j} + z'(t)\vec{k}$$

$$= (x'(t), y'(t), z'(t))$$

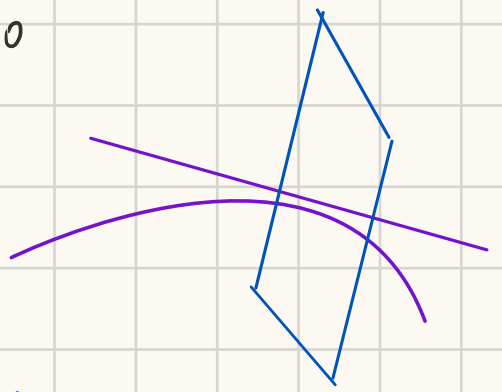
$$\vec{v}(t) = \frac{d\vec{r}}{dt} = \frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j} + \frac{dz}{dt}\vec{k}$$

$$|\vec{v}(t)| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$$

$$\frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

一、参数方程式的空间曲线的切线与法平面

$$\begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases} \quad x'(t)\ y'(t)\ z'(t) \text{ 不同时为 } 0$$



切向量 $(x'(t_0), y'(t_0), z'(t_0))$

切线方程 $\frac{x-x_0}{x'(t_0)} = \frac{y-y_0}{y'(t_0)} = \frac{z-z_0}{z'(t_0)}$

法平面平程 $x'(t_0)(x-x_0) + y'(t_0)(y-y_0) + z'(t_0)(z-z_0) = 0$

P.124 例 1 求曲线 $L: \begin{cases} x = t \\ y = -t^2 \\ z = t^3 \end{cases}$ 平行平面 $\pi: x + 2y + z = 4$ 的切线方程。

提示 平面 $\pi: \vec{n} = (1,2,1)$，曲线 L 在 t 对应点处切矢量为 $\vec{v} = (1, -2t, 3t^2)$.

切线 $\parallel$ 平面 $\pi \Rightarrow \vec{n} \cdot \vec{v} = 0 \Rightarrow 1 - 4t + 3t^2 = 0 \Rightarrow t = 1, t = \frac{1}{3}$.

当 $t = 1$ 时，对应曲线上是 $(1,-1,1)$，该点处切矢量为 $\vec{v} = (1,-2,3)$，

$\Rightarrow$ 切线方程为：$\frac{x-1}{1} = \frac{y+1}{-2} = \frac{z-1}{3}$.

当 $t = \frac{1}{3}$ 时，对应点 $\left(\frac{1}{3}, -\frac{1}{9}, \frac{1}{27}\right)$，切线矢量 $\vec{v} = \left(1, \frac{-2}{3}, \frac{1}{3}\right) \parallel (3,-2,1)$，

$\Rightarrow$ 切线方程为：$\frac{x-\frac{1}{3}}{3} = \frac{y+\frac{1}{9}}{-2} = \frac{z-\frac{1}{27}}{1}$

二、一般式方程的空间曲线的切线和法平面

$$\begin{cases} F(x,y,z) = 0 \\ G(x,y,z) = 0 \end{cases}$$ 在曲线上点 $P(x_0, y_0, z_0)$ 的切线方程

共同点 P_0 找两切平面，交线为切线

① 切平面

$F(x(t), y(t), z(t))$ 对 t 求导

$F_x'(x,y,z)x'(t) + F_y'(x,y,z)y'(t) + F_z'(x,y,z)\cdot z'(t) = 0$

代入 $x=x_0$, $y=y_0$, $z=z_0$, $t=t_0$

切线量 $\vec{l} = (x'(t_0), y'(t_0), z'(t_0))$

$\vec{n} = (F_x'(x_0,y_0,z_0), F_y'(x_0,y_0,z_0), F_z'(x_0,y_0,z_0))$

$\vec{n}\cdot\vec{l} = 0$

$\vec{n}$ 为法矢量

切平面 $F_x'(x_0,y_0,z_0)(x-x_0) + F_y'(x_0,y_0,z_0)(y-y_0) + F_z'(x_0,y_0,z_0)(z-z_0) = 0$

$F_x'(x_0,y_0,z_0)$, $F_y'(x_0,y_0,z_0)$ $F_z'(x_0,y_0,z_0)$ 不同时为0

② 切线 方程

$$\begin{cases} F_x'(P_0)(x-x_0) + F_y'(P_0)(y-y_0) + F_z'(P_0)(z-z_0) = 0 \\ G_x'(P_0)(x-x_0) + G_y'(P_0)(y-y_0) + G_z'(P_0)(z-z_0) = 0 \end{cases}$$

另一方面，直接转换为空间曲线参数方程：

曲线一般式方程 $\begin{cases} F(x,y,z)=0 \\ G(x,y,z)=0 \end{cases}$，可用方程组确定隐函数组 $\begin{cases} y=y(x) \\ z=z(x) \end{cases}$.

$$\begin{cases} y=y(x) \\ z=z(x) \end{cases} \xrightarrow[\text{参数式}]{\text{参数为}x} \begin{cases} x=x \\ y=y(x) \\ z=z(x) \end{cases} \Longrightarrow \begin{cases} F_x' + F_y'\dfrac{dy}{dx} + F_z'\dfrac{dz}{dx} = 0 \\ F_x' + F_y'\dfrac{dy}{dx} + F_z'\dfrac{dz}{dx} = 0 \end{cases} \Longrightarrow \begin{cases} \dfrac{dy}{dx} = \dfrac{\frac{\partial(F,G)}{\partial(z,x)}}{\frac{\partial(F,G)}{\partial(y,z)}} \\ \dfrac{dz}{dx} = \dfrac{\frac{\partial(F,G)}{\partial(x,y)}}{\frac{\partial(F,G)}{\partial(y,z)}} \end{cases}$$

$\Longrightarrow$ 切矢量 $\vec{l} = \left(1, \dfrac{dy}{dx}, \dfrac{dz}{dx}\right) \parallel \left(\dfrac{\partial(F,G)}{\partial(y,z)}, \dfrac{\partial(F,G)}{\partial(z,x)}, \dfrac{\partial(F,G)}{\partial(x,y)}\right)$ 求切矢量

当 F,G 在 $P_0(x_0,y_0,z_0)$ 点具有连续的偏导数，且 $\dfrac{\partial(F,G)}{\partial(y,z)}, \dfrac{\partial(F,G)}{\partial(z,x)}, \dfrac{\partial(F,G)}{\partial(x,y)}$

不同时为零时，曲线在点 $P_0(x_0,y_0,z_0)$ 处的切线方程为：

$$\frac{x-x_0}{\left.\frac{\partial(F,G)}{\partial(y,z)}\right|_{P_0}} = \frac{y-y_0}{\left.\frac{\partial(F,G)}{\partial(z,x)}\right|_{P_0}} = \frac{z-z_0}{\left.\frac{\partial(F,G)}{\partial(x,y)}\right|_{P_0}}.$$

法平面方程为：

$$\left.\frac{\partial(F,G)}{\partial(y,z)}\right|_{P_0}(x-x_0) + \left.\frac{\partial(F,G)}{\partial(z,x)}\right|_{P_0}(y-y_0) + \left.\frac{\partial(F,G)}{\partial(x,y)}\right|_{P_0}(z-z_0) = 0$$

§7.3 空间曲面的 切平面和法线

一、隐式方程空间曲面的 切平面与法向量

设过点M_0在曲面S上任作一曲线Γ: $\begin{cases} x=x(t) \\ y=y(t) \\ z=z(t) \end{cases}$，且 $\begin{cases} x_0=x(t_0) \\ y_0=y(t_0). \\ z_0=z(t_0) \end{cases}$

曲线Γ在曲面S上$\Rightarrow F(x(t),y(t),z(t))\equiv 0$，对$t$求导：

$$F_x'(x(t),y(t),z(t))x'(t)+F_y'(x(t),y(t),z(t))y'(t)+F_z'(x(t),y(t),z(t))z'(t)=0$$

把$t=t_0$代入上式得：

$$F_x'(x_0,y_0,z_0)x'(t_0)+F_y'(x_0,y_0,z_0)y'(t_0)+F_z'(x_0,y_0,z_0)z'(t_0)=0$$

$$\Leftrightarrow (F_x'(x_0,y_0,z_0),F_y'(x_0,y_0,z_0),F_z'(x_0,y_0,z_0))(x'(t_0),y'(t_0),z'(t_0))=0$$

Γ在点M_0处的切矢量：$\vec{v}\triangleq(x'(t_0),y'(t_0),z'(t_0))$

矢量$\vec{n}\triangleq(F_x'(x_0,y_0,z_0),F_y'(x_0,y_0,z_0),F_z'(x_0,y_0,z_0))$ $\Rightarrow \vec{v}\cdot\vec{n}=0$

$\Rightarrow$曲面上过M_0的任意一条曲线的切线都与矢量$\vec{n}$垂直

$\Rightarrow$证明了曲面上点M_0的任一曲线在M_0点的切线都落在以矢量$\vec{n}$为法矢量且经过M_0的平面上，这张平面称为曲面在点M_0处的切平面.

切平面的方程为：

$$F_x'(x_0,y_0,z_0)(x-x_0)+F_y'(x_0,y_0,z_0)(y-y_0)+F_z'(x_0,y_0,z_0)(z-z_0)=0$$

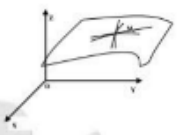
上面这个

第七节　偏导数在几何上的应用（续）

§7.3　空间曲面的切平面与法线

一、隐式方程的空间曲面的切平面与法向量

设已知曲面S方程为$F(x,y,z)=0$，$M_0(x_0,y_0,z_0)$是曲面S上一点，过点M_0在曲面S上可以作无数条曲线，这些曲线在该点分别都有切线，则这无数条切线都在同一个平面上——切平面.



设过点M_0在曲面S上任作一曲线Γ: $\begin{cases} x=x(t) \\ y=y(t) \\ z=z(t) \end{cases}$，且 $\begin{cases} x_0=x(t_0) \\ y_0=y(t_0). \\ z_0=z(t_0) \end{cases}$

曲线Γ在曲面S上$\Rightarrow F(x(t),y(t),z(t))\equiv 0$，对$t$求导：

$$F_x'(x(t),y(t),z(t))x'(t)+F_y'(x(t),y(t),z(t))y'(t)+F_z'(x(t),y(t),z(t))z'(t)=0$$

把$t=t_0$代入上式得：

$$F_x'(x_0,y_0,z_0)x'(t_0)+F_y'(x_0,y_0,z_0)y'(t_0)+F_z'(x_0,y_0,z_0)z'(t_0)=0$$

$$\Leftrightarrow (F_x'(x_0,y_0,z_0),F_y'(x_0,y_0,z_0),F_z'(x_0,y_0,z_0))(x'(t_0),y'(t_0),z'(t_0))=0$$

Γ在点M_0处的切矢量：$\vec{v}\triangleq(x'(t_0),y'(t_0),z'(t_0))$

矢量$\vec{n}\triangleq(F_x'(x_0,y_0,z_0),F_y'(x_0,y_0,z_0),F_z'(x_0,y_0,z_0))$ $\Rightarrow \vec{v}\cdot\vec{n}=0$

$\Rightarrow$曲面上过M_0的任意一条曲线的切线都与矢量$\vec{n}$垂直

$\Rightarrow$证明了曲面上点M_0的任一曲线在M_0点的切线都落在以矢量$\vec{n}$为法矢量且经过M_0的平面上，这张平面称为曲面在点M_0处的切平面.

切平面的方程为：

$$F_x'(x_0,y_0,z_0)(x-x_0)+F_y'(x_0,y_0,z_0)(y-y_0)+F_z'(x_0,y_0,z_0)(z-z_0)=0$$

称曲面在点M_0处切平面的法矢量为曲面在点M_0的法线矢量（法矢量）.过点M_0垂直于曲面在该点切平面的直线，称为曲面在点M_0的法线.

由于该线经过M_0点，方向向量平行$\vec{n}$，因此法线的方程为

$$\frac{x-x_0}{F_x'(x_0,y_0,z_0)}=\frac{y-y_0}{F_y'(x_0,y_0,z_0)}=\frac{z-z_0}{F_z'(x_0,y_0,z_0)}$$

曲面在点M_0处存在切平面的充分条件：F_x',F_y',F_z'在M_0点具有连续的偏导数且$F_x'(x_0,y_0,z_0),F_y'(x_0,y_0,z_0),F_z'(x_0,y_0,z_0)$不同时为零

P.126 例 2　求曲面$x^2+y^2+z^2-xy-3=0$的切平面，使该切平面同时垂直于平面$\pi_1: z=0$与$\pi_2: x+y+z+1=0$.

详见 P.126

提示：切方向与平面π_1、平面π_2的两个法矢量垂直+$x^2+y^2+z^2-xy-3=0$,可以求得切点。

P.127 例 3　证明曲面$S: F(nx-lz,ny-mz)=0$在任意一点处的切平面都平行于直线$L: \frac{x-1}{l}=\frac{y-2}{m}=\frac{z-3}{n}$

详见 P.127

提示：曲面$S: F(nx-lz,ny-mz)=0$一定要转换成标准形式！

补充题 1：求椭球面$S: \frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ 在点$P_0(x_0,y_0,z_0)$处的切平面.

解：$F(x,y,z)=\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1$，则椭球面的法向量

$$\vec{n}=(F_x',F_y',F_z')=\left(\frac{2x}{a^2},\frac{2y}{b^2},\frac{2z}{c^2}\right)\Big|_{P_0}=\left(\frac{2x_0}{a^2},\frac{2y_0}{b^2},\frac{2z_0}{c^2}\right)$$

切平面方程 π: $\frac{x_0}{a^2}(x-x_0)+\frac{y_0}{b^2}(y-y_0)+\frac{z_0}{c^2}(z-z_0)=0$

$$\Leftrightarrow \pi: \frac{x_0x}{a^2}+\frac{y_0y}{b^2}+\frac{z_0z}{c^2}=1$$

特殊地：

球面$x^2+y^2+z^2=r^2(r>0)$ 在点$P_0(x_0,y_0,z_0)$处的切平面：

$$\pi: x_0x+y_0y+z_0z=r^2$$

$F(x,y,z)=0$　法矢量 $(F_x'(x_0,y_0,z_0)\ F_y'(x_0,y_0,z_0)\ F_z'(x_0,y_0,z_0))$

显式 $z=f(x,y)$　法矢量 $(f_x'(x,y),f_y'(x,y),-1)$

$\Rightarrow$曲面在点P_0处的切平面方程:

$$f_x'(x_0,y_0)(x-x_0)+f_y'(x_0,y_0)(y-y_0)-(z-z_0)=0$$

法线方程: $\dfrac{x-x_0}{f_x'(x_0,y_0)}=\dfrac{y-y_0}{f_y'(x_0,y_0)}=\dfrac{z-z_0}{-1}$

⚠ 也可利用偏导数的几何意义推得。

补充题 2 设$z=f(x,y)$在R^2上连续，且$\lim\limits_{\substack{x\to0\\y\to0}}\dfrac{f(x,y)-x+3y-4}{x^2+y^2}=1$,

求曲面$z=f(x,y)$在点$(0,0)$处的切平面方程。

解: $\begin{cases}z=f(x,y)\text{在点}(0,0)\text{处连续}\\ \lim\limits_{\substack{x\to0\\y\to0}}\dfrac{f(x,y)-x+3y-4}{x^2+y^2}=1\end{cases}$

$$\Rightarrow\begin{cases}f(0,0)=\lim\limits_{\substack{x\to0\\y\to0}}f(x,y)=4\ (\text{切点为}(0,0,4))\\ f(x,y)=4+x-3y+x^2+y^2+o(\rho^2)\quad(\rho=\sqrt{x^2+y^2})\end{cases}$$

$$\Rightarrow f(x,y)=f(0,0)+x-3y+o(\rho)\Rightarrow\begin{cases}z=f(x,y)\text{在点}(0,0)\text{处连续}\\ dz|_{(0,0)}=dx-3dy\end{cases}$$

$$\Rightarrow\begin{cases}f_x'(0,0)=1\\ f_y'(0,0)=-3\end{cases}\Rightarrow\text{在点}(0,0)\text{处的切平面}\pi:z=x-3y+4.$$

三、参数式方程的空间曲面的切平面与法向量

设曲面参数方程$\begin{cases}x=x(u,v)\\ y=y(u,v)\\ z=z(u,v)\end{cases}$在$(u_0,v_0)$对应曲面上点$P_0(x_0,y_0,z_0)$

解: $\begin{cases}x=x(u,v)\\ y=y(u,v)\end{cases}$可确定反函数组$\begin{cases}u=u(x,y)\\ v=v(x,y)\end{cases}$ 代入$z=z(u,v)$

$\Leftrightarrow z=z[u(x,y),v(x,y)]\Leftrightarrow z$是$x,y$的函数.

$\Rightarrow z=z(u,v)$可看成z通过中间变量x,y是u,v的复合函数

由复合函数求导法则知:

$$\begin{cases}\dfrac{\partial z}{\partial u}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial u}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial u}\\ \dfrac{\partial z}{\partial v}=\dfrac{\partial z}{\partial x}\dfrac{\partial x}{\partial v}+\dfrac{\partial z}{\partial y}\dfrac{\partial y}{\partial v}\end{cases}\Rightarrow\dfrac{\partial z}{\partial x}=\dfrac{\dfrac{\partial(z,y)}{\partial(u,v)}}{\dfrac{\partial(x,y)}{\partial(u,v)}}\quad\text{同理:}\quad\dfrac{\partial z}{\partial y}=\dfrac{\dfrac{\partial(x,z)}{\partial(u,v)}}{\dfrac{\partial(x,y)}{\partial(u,v)}}.$$

(或者，利用参数方程直接求出用dx,dy来表示dz)

$$F(x,y,z)\triangleq z[u(x,y),v(x,y)]-z\Rightarrow\begin{cases}F_x'=\dfrac{\partial z}{\partial x}\\ F_y'=\dfrac{\partial z}{\partial y}\\ F_z'=-1\end{cases}\Rightarrow\text{曲面点}P_0\text{处的法矢量}\vec{n}$$

$$\vec{n}=\left(\frac{\partial z}{\partial x},\frac{\partial z}{\partial y},-1\right)\Big|_{(u_0,v_0)}=\left(\frac{\dfrac{\partial(z,y)}{\partial(u,v)}}{\dfrac{\partial(x,y)}{\partial(u,v)}},\frac{\dfrac{\partial(x,z)}{\partial(u,v)}}{\dfrac{\partial(x,y)}{\partial(u,v)}},-1\right)\Big|_{(u_0,v_0)}$$

$$\parallel\left(\frac{\partial(y,z)}{\partial(u,v)},\frac{\partial(z,x)}{\partial(u,v)},\frac{\partial(x,y)}{\partial(u,v)}\right)\Big|_{(u_0,v_0)}$$

记:

$$A\triangleq\frac{\partial(y,z)}{\partial(u,v)}\Big|_{(u_0,v_0)},B\triangleq\frac{\partial(z,x)}{\partial(u,v)}\Big|_{(u_0,v_0)},C\triangleq\frac{\partial(x,y)}{\partial(u,v)}\Big|_{(u_0,v_0)}$$

$\Rightarrow$该曲面在点$P_0(x_0,y_0,z_0)$处的切平面方程:

$$A(x-x_0)+B(y-y_0)+C(z-z_0)=0$$

法线方程: $\dfrac{x-x_0}{A}=\dfrac{y-y_0}{B}=\dfrac{z-z_0}{C}$

上述分析得: 曲面$\begin{cases}x=x(u,v)\\ y=y(u,v)\\ z=z(u,v)\end{cases}$在$(u_0,v_0)$对应点$P_0$切平面存在的充分条件:

$x(u,v),y(u,v),z(u,v)$具有连续的偏导数且$\dfrac{\partial(y,z)}{\partial(u,v)}\Big|_{(u_0,v_0)},\dfrac{\partial(z,x)}{\partial(u,v)}\Big|_{(u_0,v_0)},\dfrac{\partial(x,y)}{\partial(u,v)}\Big|_{(u_0,v_0)}$不同时为零。

⚠ 也可利用偏导数的几何意义推得。

例如 求曲面$\begin{cases}x=u\cos v\\ y=u\sin v\\ z=2v\end{cases}$在$\begin{cases}u=1\\ v=\frac{\pi}{4}\end{cases}$所对应点处的切平面方程与法线方程.

解: 方法一 直接利用公式:

$$\begin{cases}x=u\cos v\\ y=u\sin v\\ z=2v\end{cases}\Rightarrow\begin{cases}\dfrac{\partial x}{\partial u}=\cos v\\ \dfrac{\partial x}{\partial v}=-u\sin v\end{cases},\begin{cases}\dfrac{\partial y}{\partial u}=\sin v\\ \dfrac{\partial y}{\partial v}=u\cos v\end{cases},\begin{cases}\dfrac{\partial z}{\partial u}=0\\ \dfrac{\partial z}{\partial v}=2\end{cases}.$$

$$\frac{\partial(y,z)}{\partial(u,v)}\Big|_{(1,\frac{\pi}{4})}=\begin{vmatrix}\frac{\sqrt2}{2}&\frac{\sqrt2}{2}\\ 0&2\end{vmatrix}=\sqrt2,\quad\frac{\partial(z,x)}{\partial(u,v)}\Big|_{(1,\frac{\pi}{4})}=\begin{vmatrix}0&2\\ \frac{\sqrt2}{2}&-\frac{\sqrt2}{2}\end{vmatrix}=-\sqrt2,$$

$$\frac{\partial(x,y)}{\partial(u,v)}\Big|_{(1,\frac{\pi}{4})}=\begin{vmatrix}\frac{\sqrt2}{2}&-\frac{\sqrt2}{2}\\ \frac{\sqrt2}{2}&\frac{\sqrt2}{2}\end{vmatrix}=1,$$

所求的切平面方程:

$$\sqrt2\left(x-\frac{\sqrt2}{2}\right)-\sqrt2\left(y-\frac{\sqrt2}{2}\right)+\left(z-\frac{\pi}{2}\right)=0\Leftrightarrow\sqrt2x-\sqrt2y+z=\frac{\pi}{2}$$

法线方程: $\dfrac{x-\frac{\sqrt2}{2}}{\sqrt2}=\dfrac{y-\frac{\sqrt2}{2}}{-\sqrt2}=\dfrac{z-\frac{\pi}{2}}{1}.$

法二: 求出z_x',z_y',得切平面的法矢量为$\vec{n}=(z_x',z_y',-1)$

$$\begin{cases}x=u\cos v\\ y=u\sin v\end{cases}\Rightarrow\begin{cases}dx=\cos v\,du-u\sin v\,dv\\ dy=\sin v\,du+u\cos v\,dv\end{cases}\Rightarrow\begin{cases}du=\cos v\,dx+\sin v\,dy.\\ dv=\dfrac{1}{u}(\cos v\,dy-\sin v\,dx)\end{cases}$$

$$z=2v\Rightarrow dz=2dv=\frac{2}{u}(\cos v\,dy-\sin v\,dx)$$

$$\Rightarrow\vec{v}=\left(-\frac{2\sin v}{u},\frac{2\cos v}{u},-1\right)\Big|_{(1,\frac{\pi}{4})}=(-\sqrt2,\sqrt2,-1)\parallel(\sqrt2,-\sqrt2,1)$$

所求的切平面方程:

$$\sqrt2\left(x-\frac{\sqrt2}{2}\right)-\sqrt2\left(y-\frac{\sqrt2}{2}\right)+\left(z-\frac{\pi}{2}\right)=0\Leftrightarrow\sqrt2x-\sqrt2y+z=\frac{\pi}{2}$$

法线方程: $\dfrac{x-\frac{\sqrt2}{2}}{\sqrt2}=\dfrac{y-\frac{\sqrt2}{2}}{-\sqrt2}=\dfrac{z-\frac{\pi}{2}}{1}.$

法三: 利用偏导数的几何意义

曲线$\begin{cases}C_1:u=1\\ C_2:v=\frac{\pi}{4}\end{cases}\Rightarrow$切向量$\begin{cases}\vec{v}_1=(x_v'(u_0,v_0),y_v'(u_0,v_0),z_v'(u_0,v_0))\\ \vec{v}_2=(x_u'(u_0,v_0),y_u'(u_0,v_0),z_u'(u_0,v_0))\end{cases}$

$$\begin{cases}x=u\cos v\\ y=u\sin v\\ z=2v\end{cases}\Rightarrow\begin{cases}x_u'=\cos v\\ y_u'=\sin v,\\ z_u'=0\end{cases}\begin{cases}x_v'=-u\sin v\\ y_v'=u\cos v\\ z_v'=2v\end{cases}\Rightarrow\begin{matrix}\vec{v}_1=\left(-\frac{\sqrt2}{2},\frac{\sqrt2}{2},2\right)\\ \vec{v}_2=\left(\frac{\sqrt2}{2},\frac{\sqrt2}{2},0\right)\end{matrix}$$

$$\vec{n}=\vec{v}_2\times\vec{v}_1=\begin{vmatrix}\vec{i}&\vec{j}&\vec{k}\\ -\frac{\sqrt2}{2}&\frac{\sqrt2}{2}&2\\ \frac{\sqrt2}{2}&\frac{\sqrt2}{2}&0\end{vmatrix}=-\sqrt2\vec{i}+\sqrt2\vec{j}-\vec{k}.$$

$\Rightarrow$该曲面在点$P_0(x_0,y_0,z_0)$处的切平面和法线方程:

$$-\sqrt2\left(x-\frac{\sqrt2}{2}\right)+\sqrt2\left(y-\frac{\sqrt2}{2}\right)-\left(z-\frac{\pi}{2}\right)=0\Leftrightarrow\sqrt2x-\sqrt2y+z=\frac{\pi}{2}$$

$$\frac{x-\frac{\sqrt2}{2}}{\sqrt2}=\frac{y-\frac{\sqrt2}{2}}{-\sqrt2}=\frac{z-\frac{\pi}{2}}{1}.$$

全微分、泰勒展开与切平面之间的联系:

设$z=f(x,y)$在点$P_0(x_0,y_0)$处存在一阶连续偏导数，则:

一阶Taylor展开(全增量公式)为: $(\rho\to0)$

$$f(x,y)=f(x_0,y_0)+f_x'(x_0,y_0)(x-x_0)+f_y'(x_0,y_0)(y-y_0)+o(\rho)$$

$$\Leftrightarrow\Delta z\triangleq f(x,y)-f(x_0,y_0)=f_x'(x_0,y_0)(x-x_0)+f_y'(x_0,y_0)(y-y_0)+o(\rho)$$

$\Leftrightarrow z=f(x,y)$在点$P_0(x_0,y_0)$处的全微分.

$\Rightarrow z - z_0 = f_x'(x_0, y_0)(x - x_0) + f_y'(x_0, y_0)(y - y_0)$

——光滑曲面$S: z = f(x, y)$在点$P_0(x_0, y_0)$处的切平面方程.

补充题 3：设函数$F(x, y, z)$具有一阶连续偏导，

证明：曲面$F\left(\frac{z}{y}, \frac{x}{z}, \frac{y}{x}\right) = 0$上任何一点的切平面均过定点。

证明：$g(x, y, z) \triangleq F\left(\frac{z}{y}, \frac{x}{z}, \frac{y}{x}\right) \Rightarrow \begin{cases} g_x' = \frac{1}{z}F_2' - \frac{y}{x^2}F_3' \\ g_y' = -\frac{1}{y^2}F_1' + \frac{1}{x}F_3' \\ g_z' = \frac{1}{y}F_1' - \frac{x}{z^2}F_2' \end{cases}$

$\Rightarrow$ 曲面$F\left(\frac{z}{y}, \frac{x}{z}, \frac{y}{x}\right) = 0$在任一点$(x^*, y^*, z^*)$的切平面方程为：

$$\left(\frac{1}{z^*}F_2' - \frac{y^*}{(x^*)^2}F_3'\right)(x - x^*) + \left(-\frac{z^*}{(y^*)^2}F_1' + \frac{1}{x^*}F_3'\right)(y - y^*) + \left(\frac{1}{y^*}F_1' - \frac{x^*}{(z^*)^2}F_2'\right)(z - z^*) = 0 \quad (*)$$

$x = 0, y = 0, z = 0$代入切平面方程$(*)$得：

$$-\left(\frac{1}{z^*}F_2' - \frac{y^*}{(x^*)^2}F_3'\right)x^* - \left(-\frac{z^*}{(y^*)^2}F_1' + \frac{1}{x^*}F_3'\right)y^* - \left(\frac{1}{y^*}F_1' - \frac{x^*}{(z^*)^2}F_2'\right)z^*$$

$$\Leftrightarrow -\left(\frac{x^*}{z^*}F_2' - \frac{y^*}{x^*}F_3'\right) + \left(\frac{z^*}{y^*}F_1' - \frac{y^*}{x^*}F_3'\right) - \left(\frac{z^*}{y^*}F_1' - \frac{x^*}{z^*}F_2'\right) = 0.$$

$\Rightarrow$点$(0,0,0)$满足切平面方程$(*) \Rightarrow$ 切平面方程$(*)$过原点$(0,0,0)$.

注：切平面方程$(*)$$F_1', F_2', F_3'$均在$(x^*, y^*, z^*)$处的$F_1', F_2', F_3'$.

习题定点